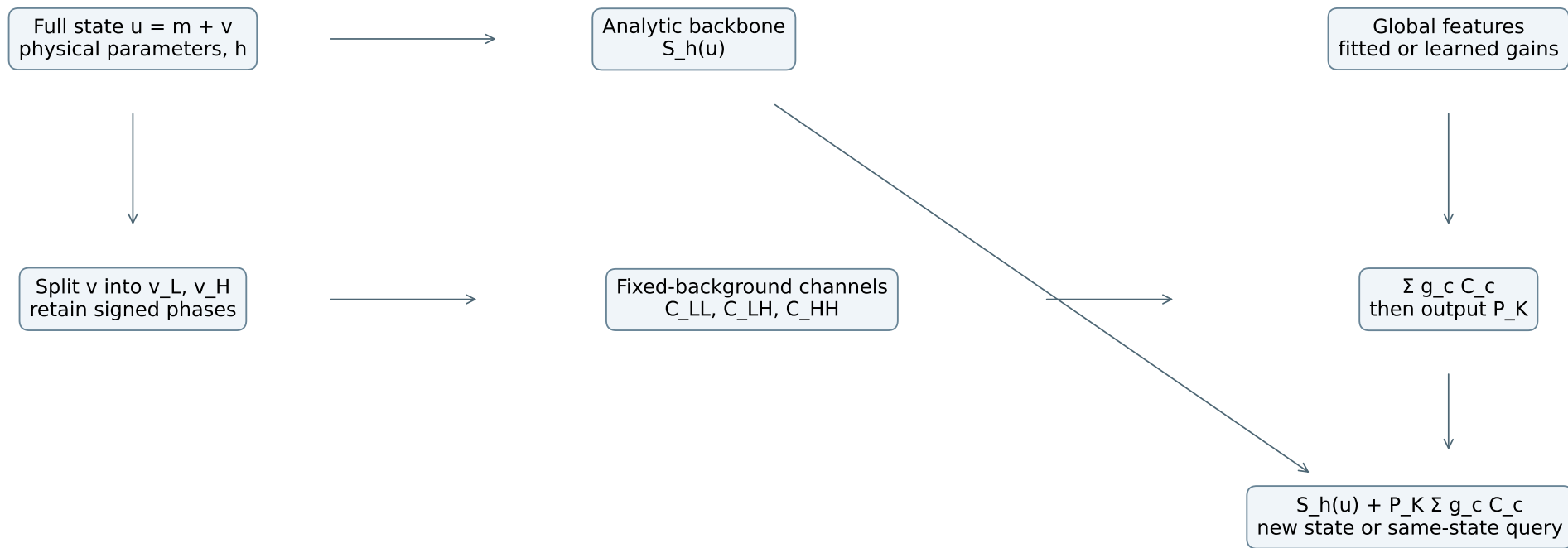
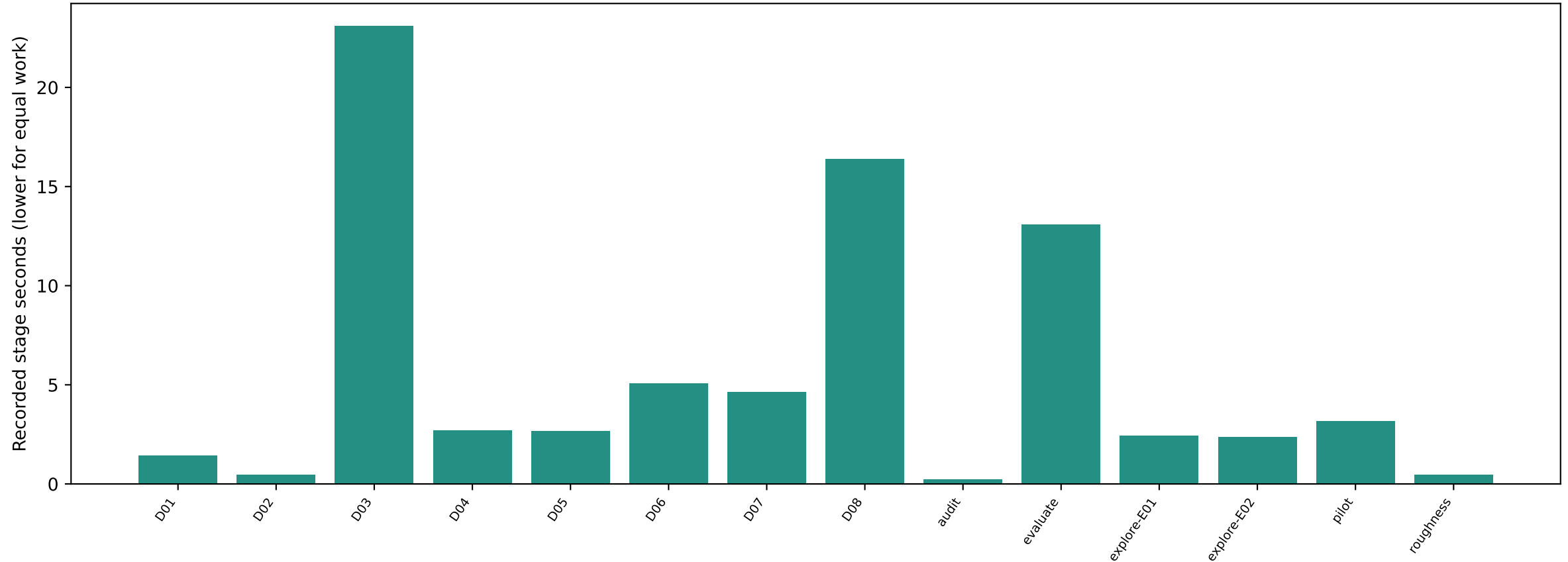


## Architecture and information flow



READ: Arrows denote data flow, not better/worse. Input-origin channels are not a third spatial dimension. Equal unit gains reproduce normalized quadratic correction.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Fixed physical background and operator shared by every channel. Learning receives features; the physical branch still receives the full field. Promotion into main TDN requires a separate decision.

## Stage completion and observed compute



READ: Y lower is less stage time for equal completed work. A short failed stage is not a scientific win. Missing measurements are NA; walltime ceilings are not consumed costs.

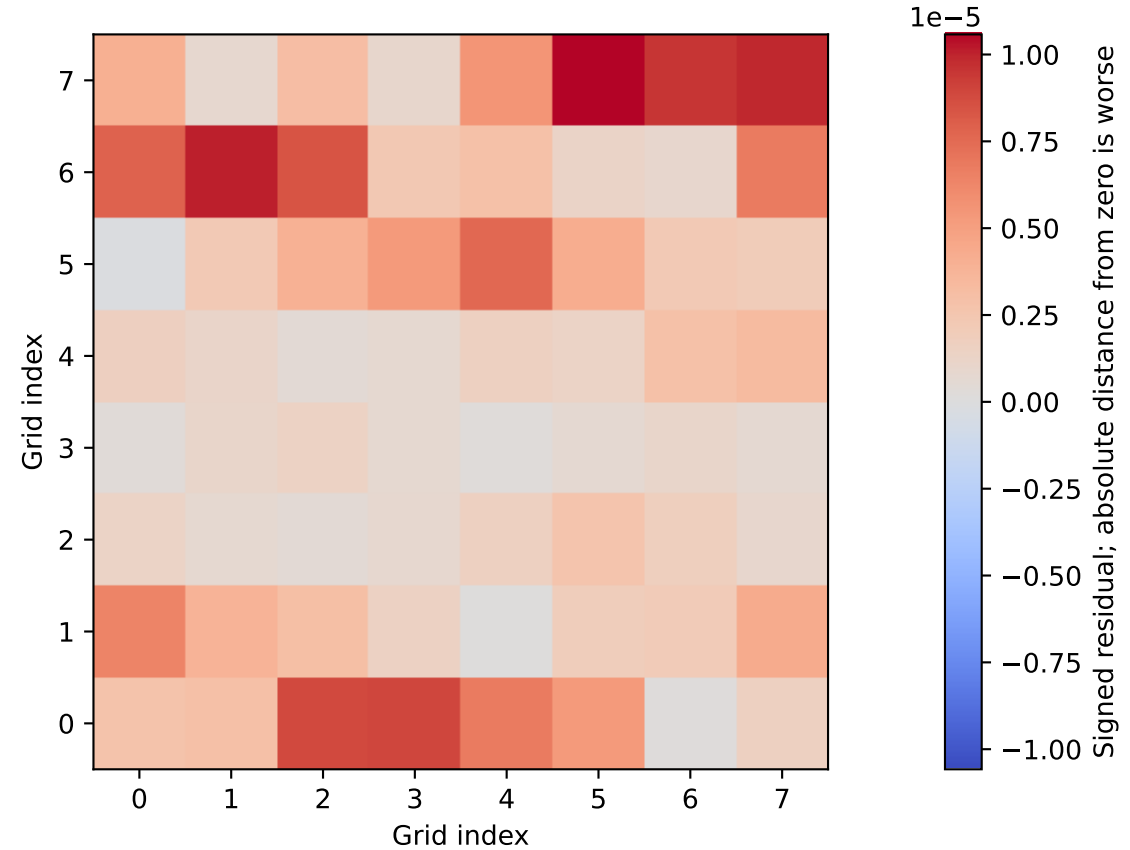
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spatial residual Theirs: df [discrete]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

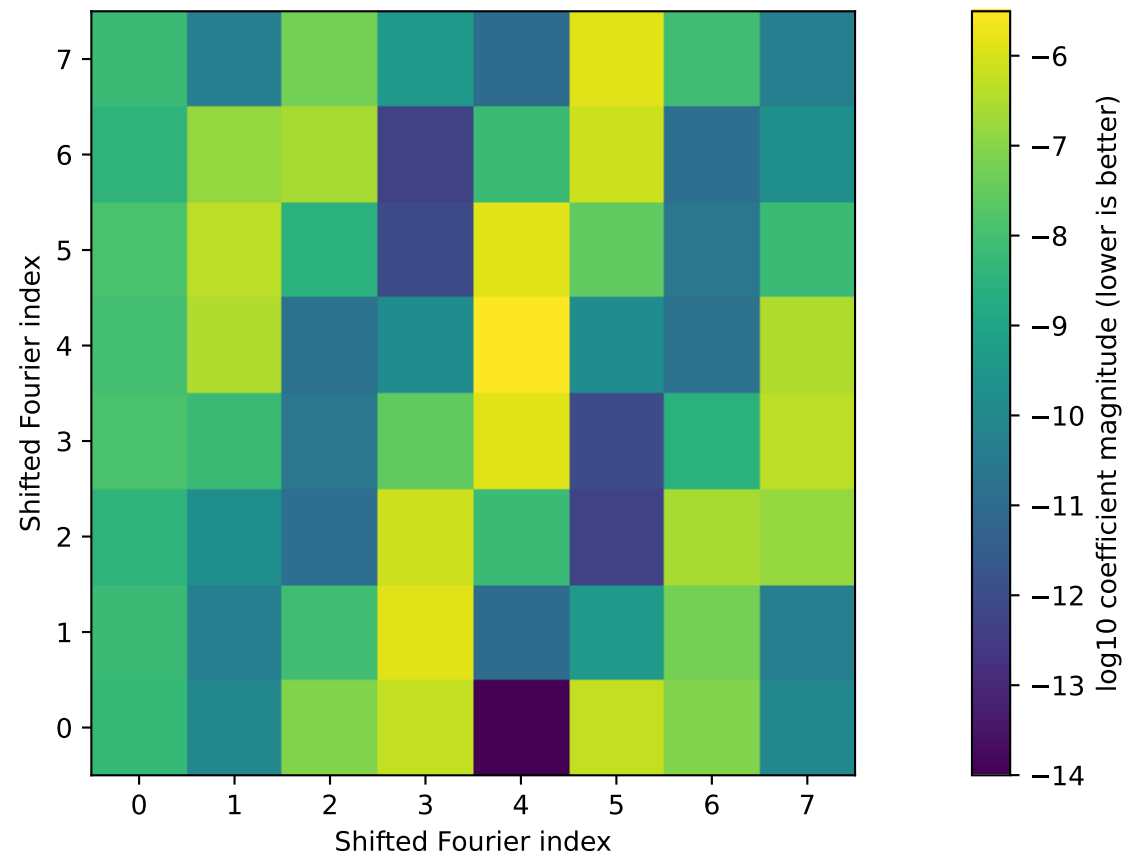
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## D01 Spectral residual Theirs: df [discrete]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

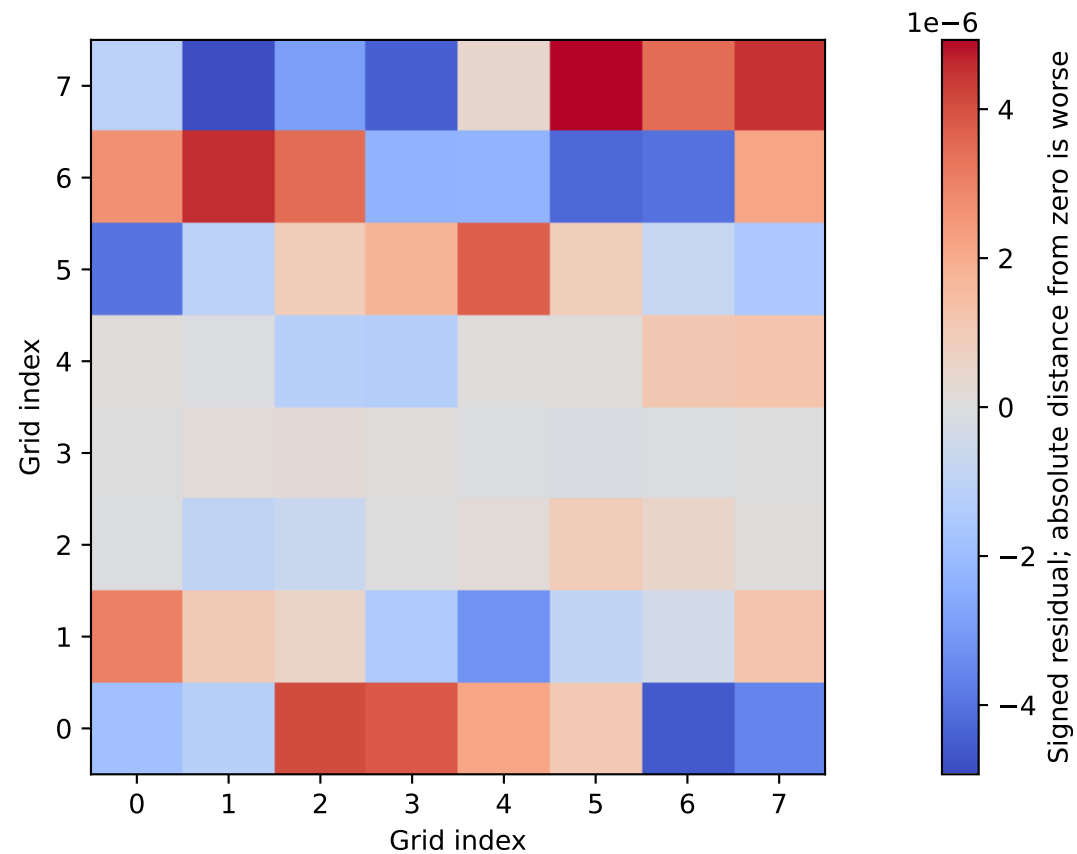
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## D01 Spatial residual Analytic control: quad2\_fixed [discrete]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

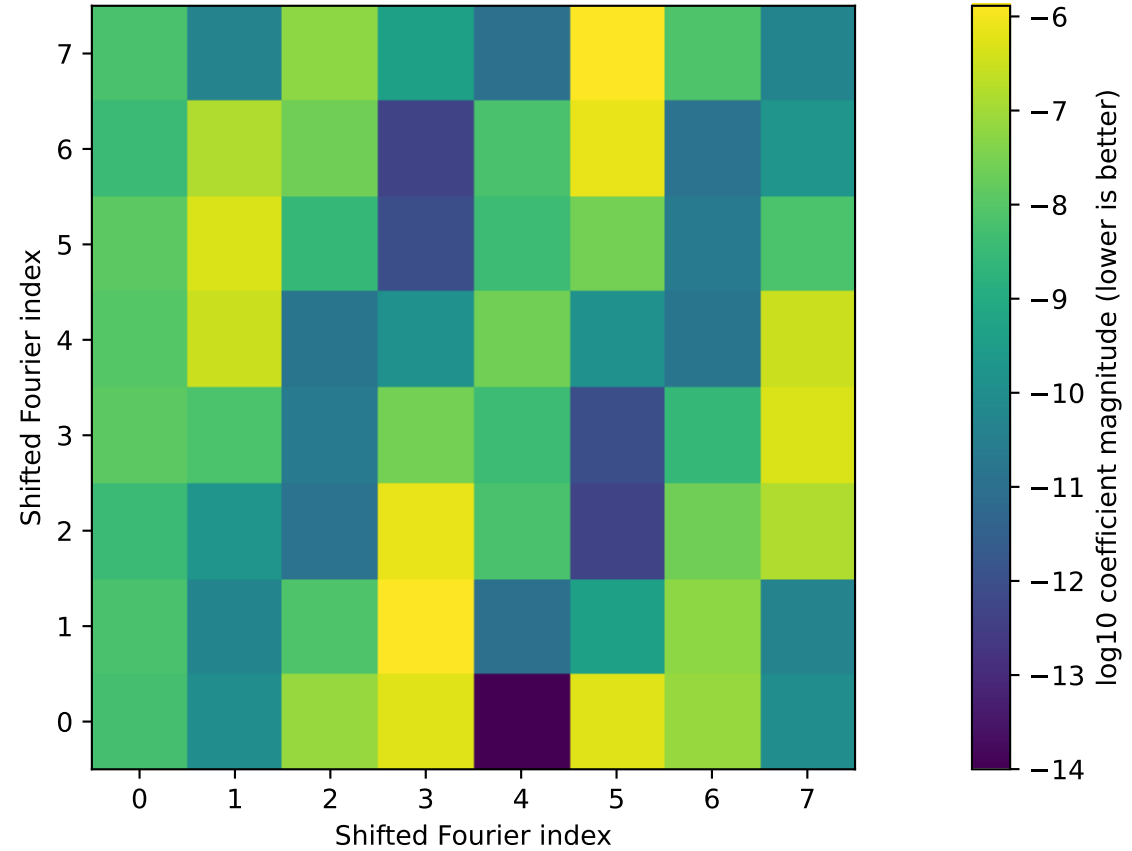
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: quad2\_fixed [discrete]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

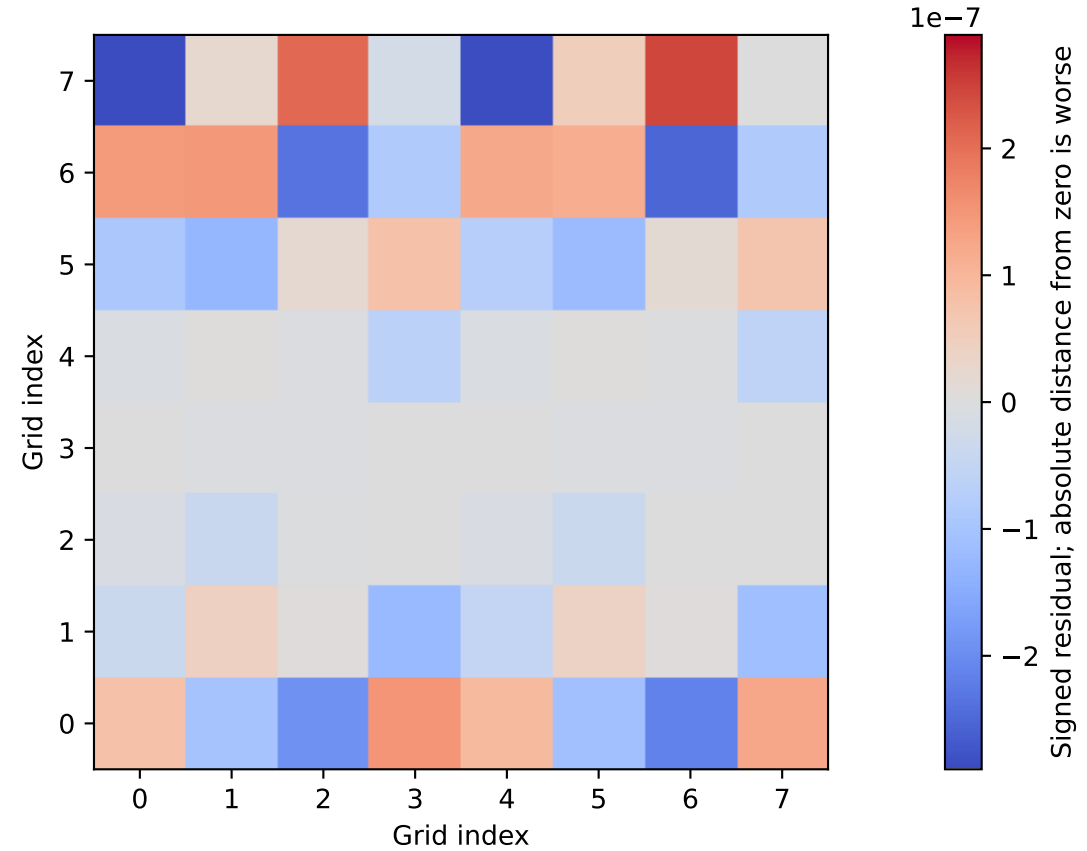
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spatial residual Analytic control: analytic\_quad\_cubic [discrete]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

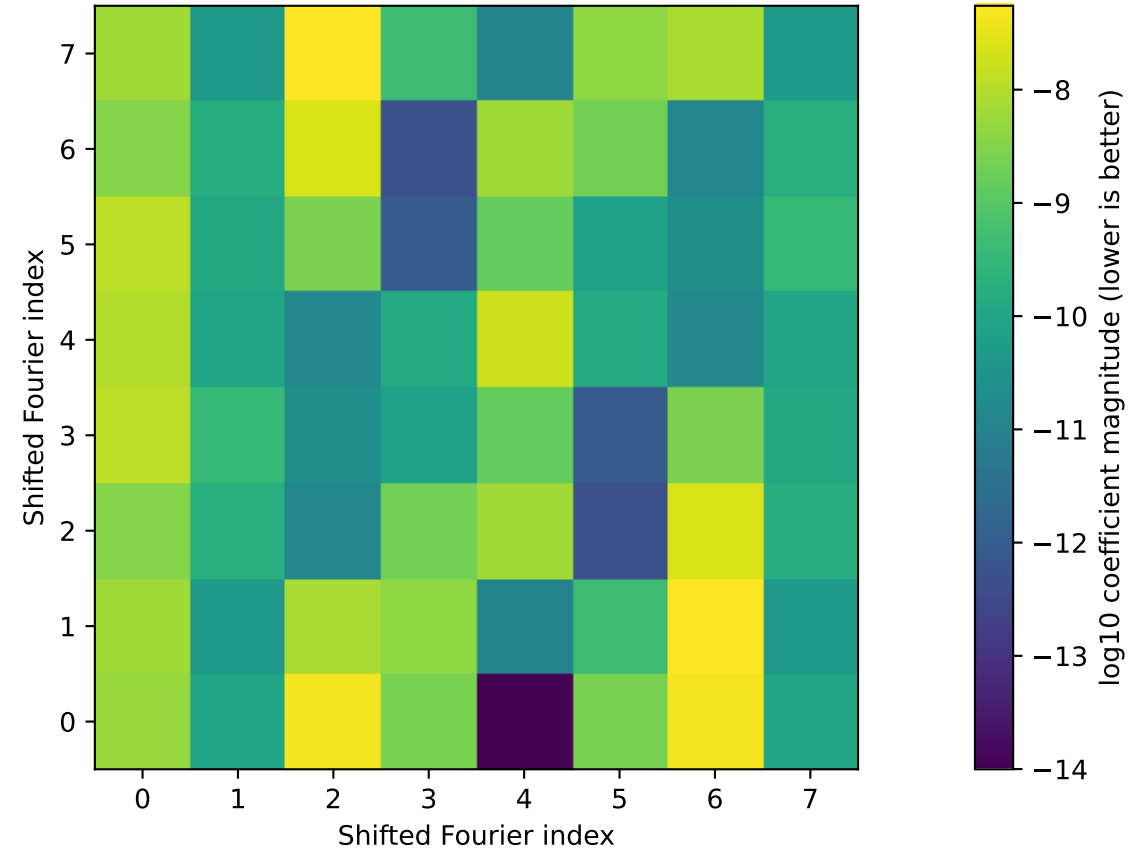
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: analytic\_quad\_cubic [discrete]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

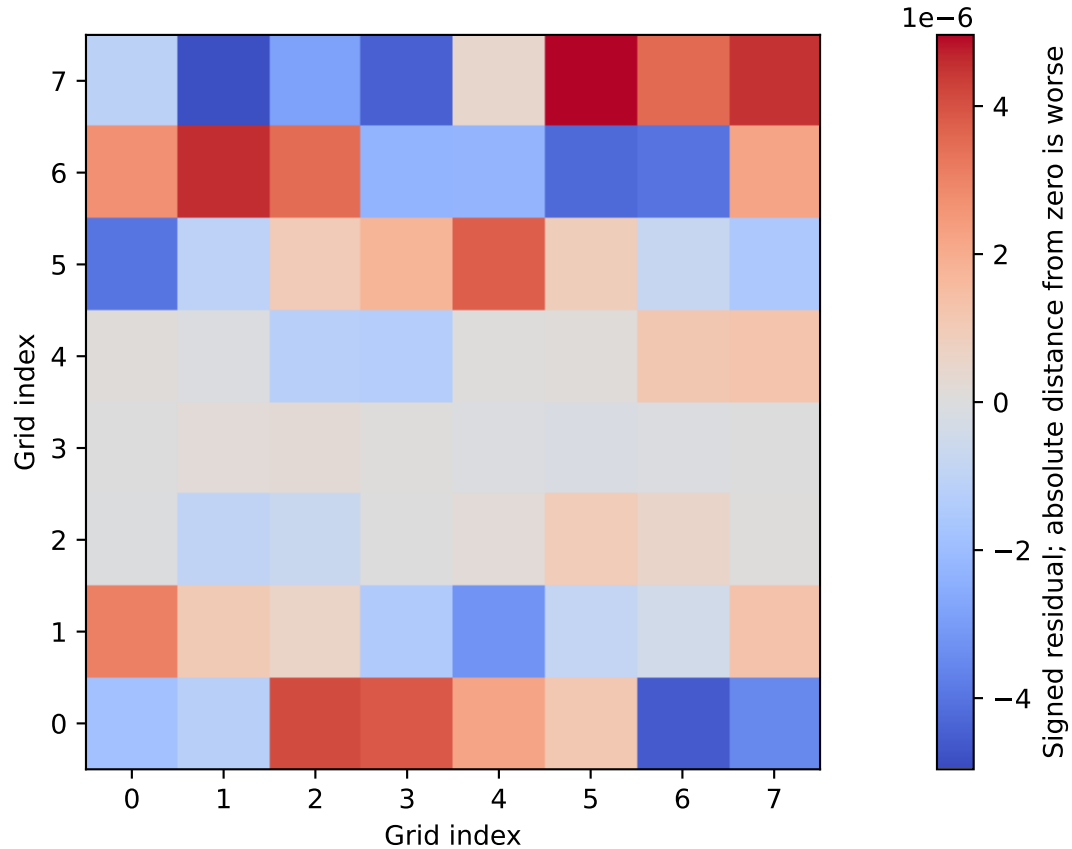
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

D01 Spatial residual Analytic control: scalar\_bounded\_oracle [discrete]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

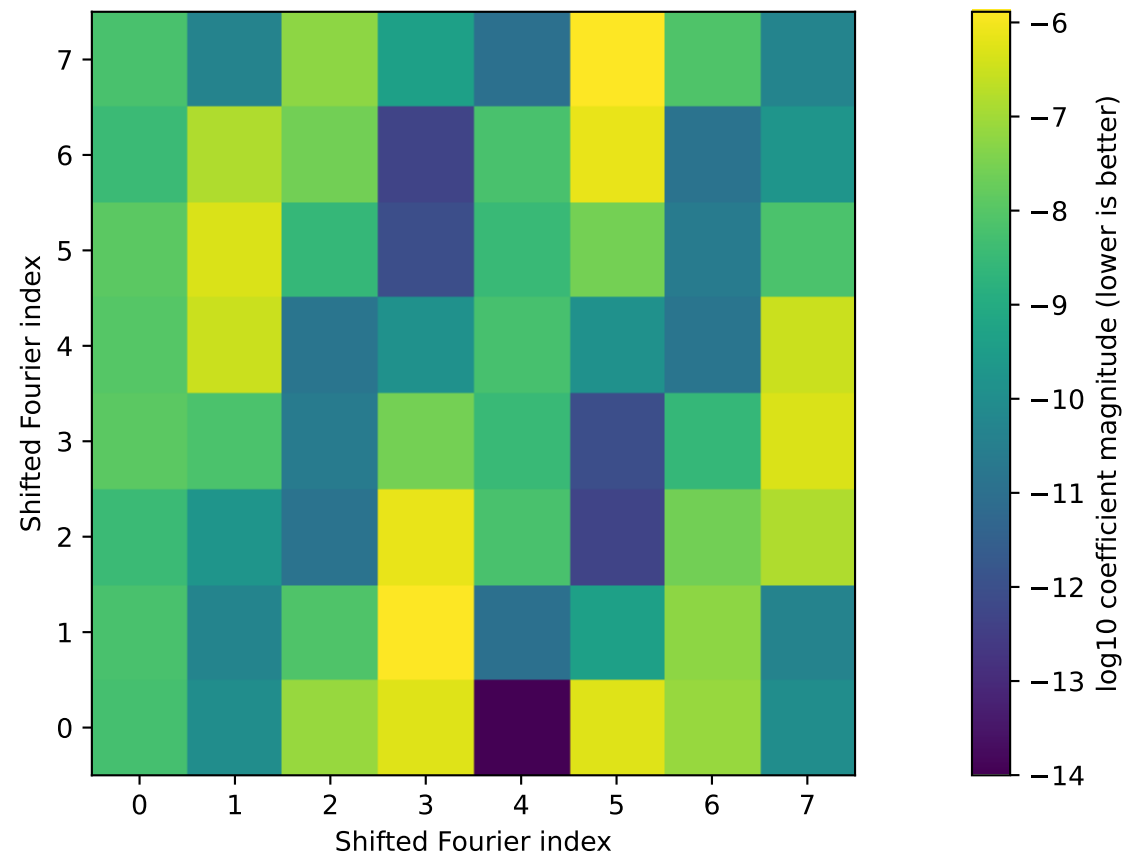
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: scalar\_bounded\_oracle [discrete]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

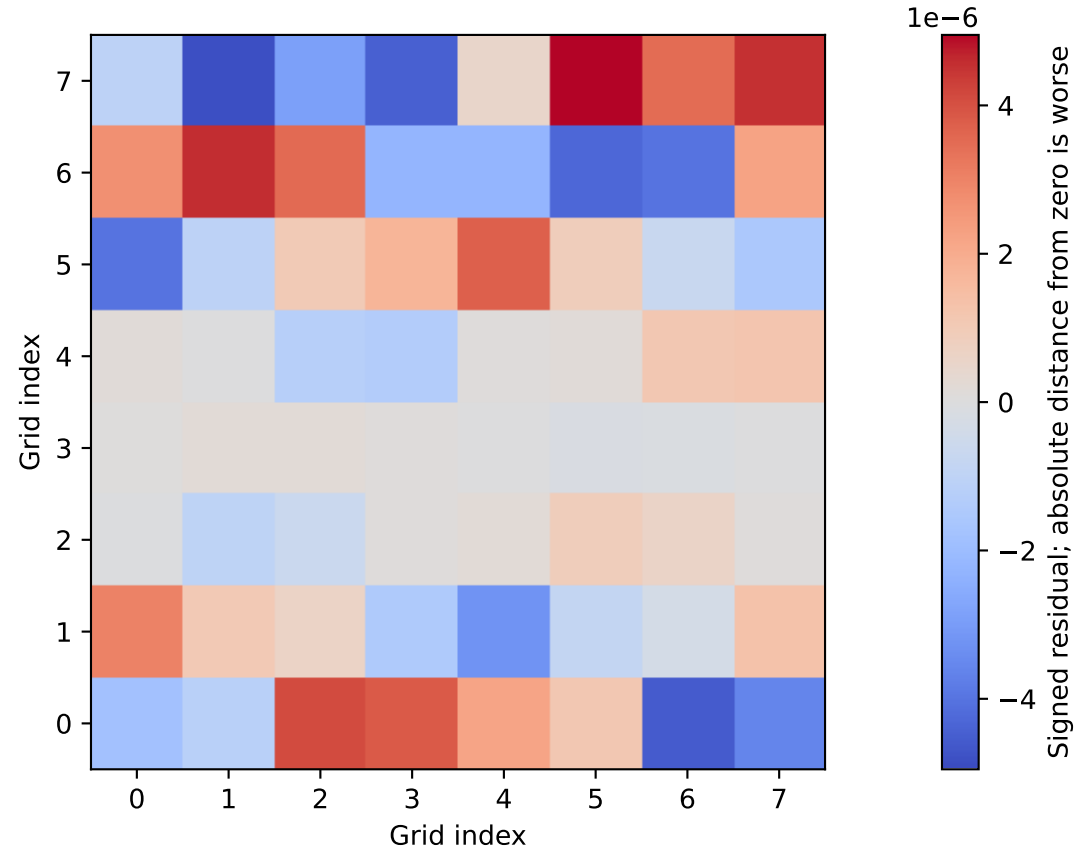
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spatial residual Analytic control: interaction\_channel\_bounded\_oracle [discrete]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

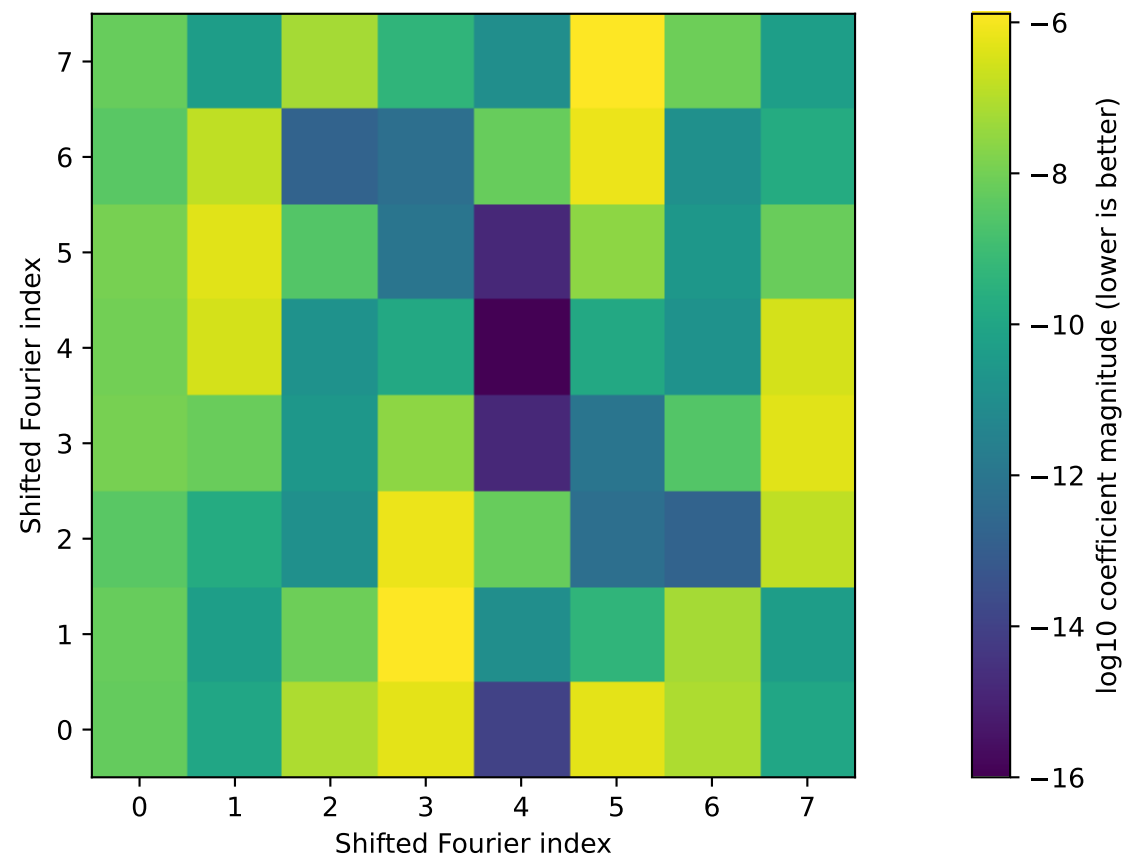
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## D01 Spectral residual Analytic control: interaction\_channel\_bounded\_oracle [discrete]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

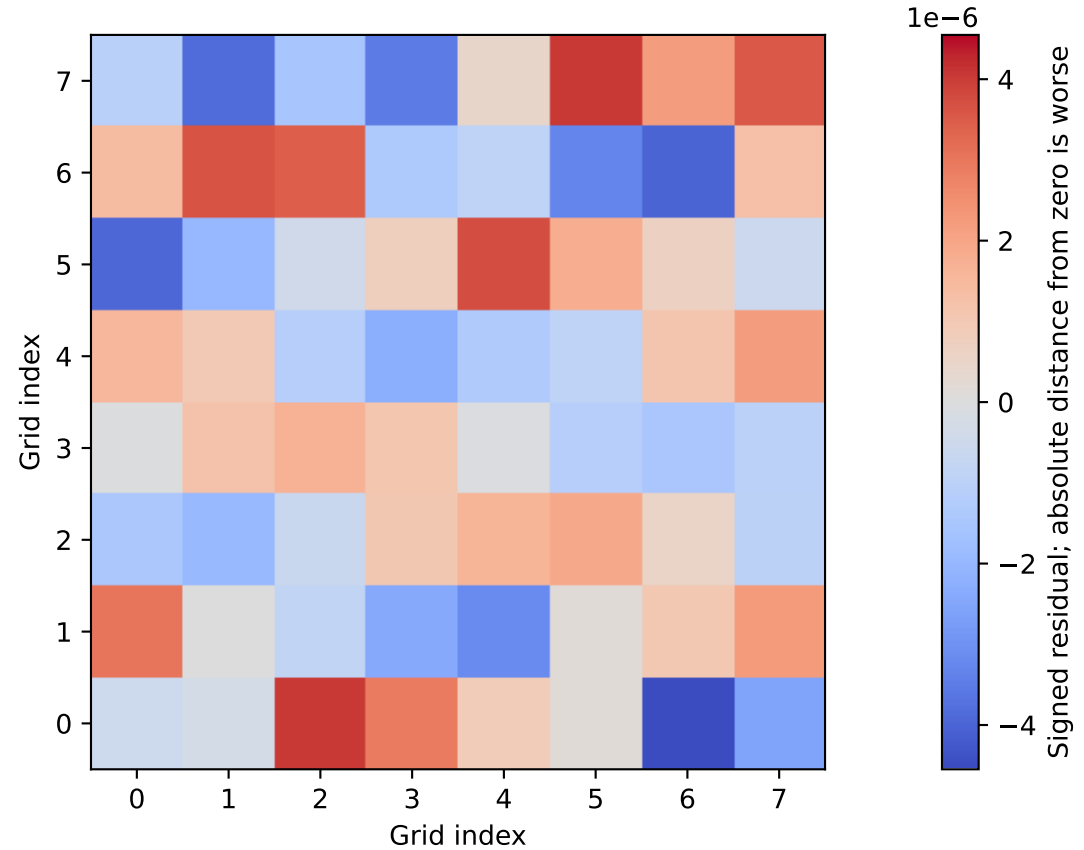
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## D01 Spatial residual Analytic control: quadratic\_cubic\_bounded\_oracle [discrete]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

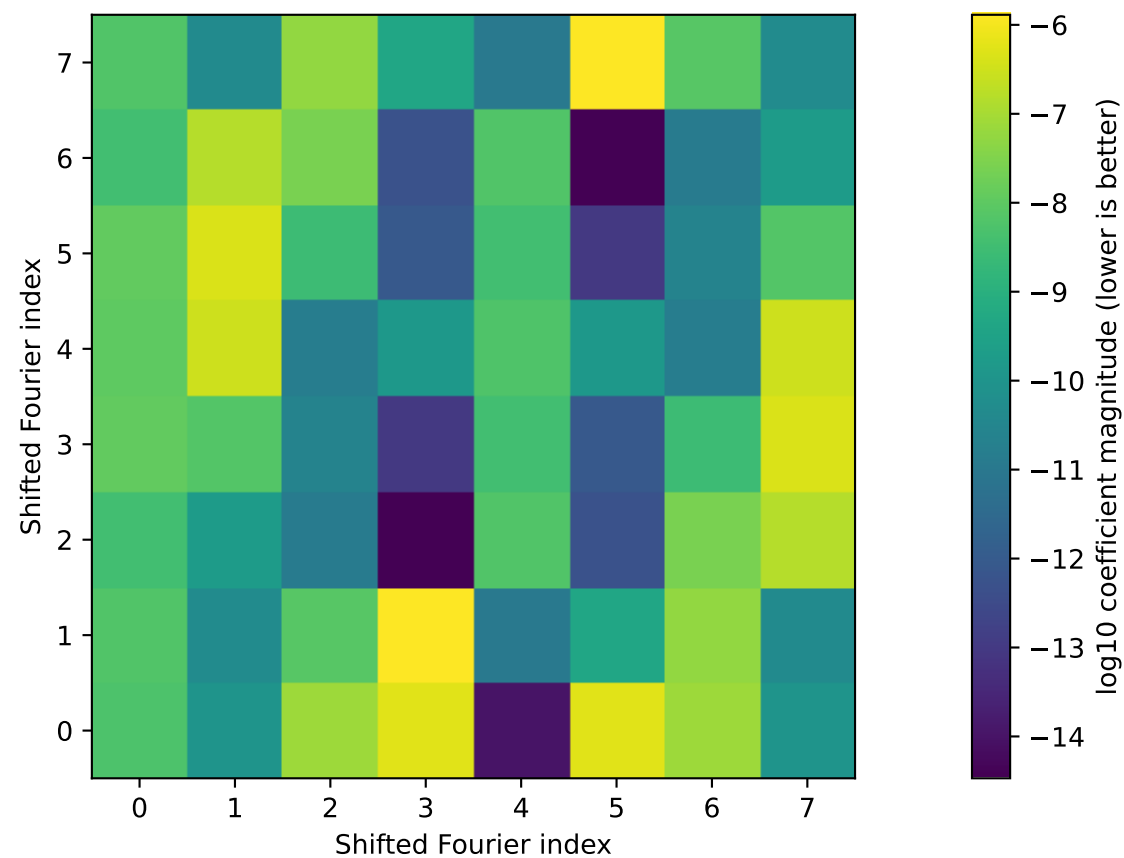
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: quadratic\_cubic\_bounded\_oracle [discrete]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

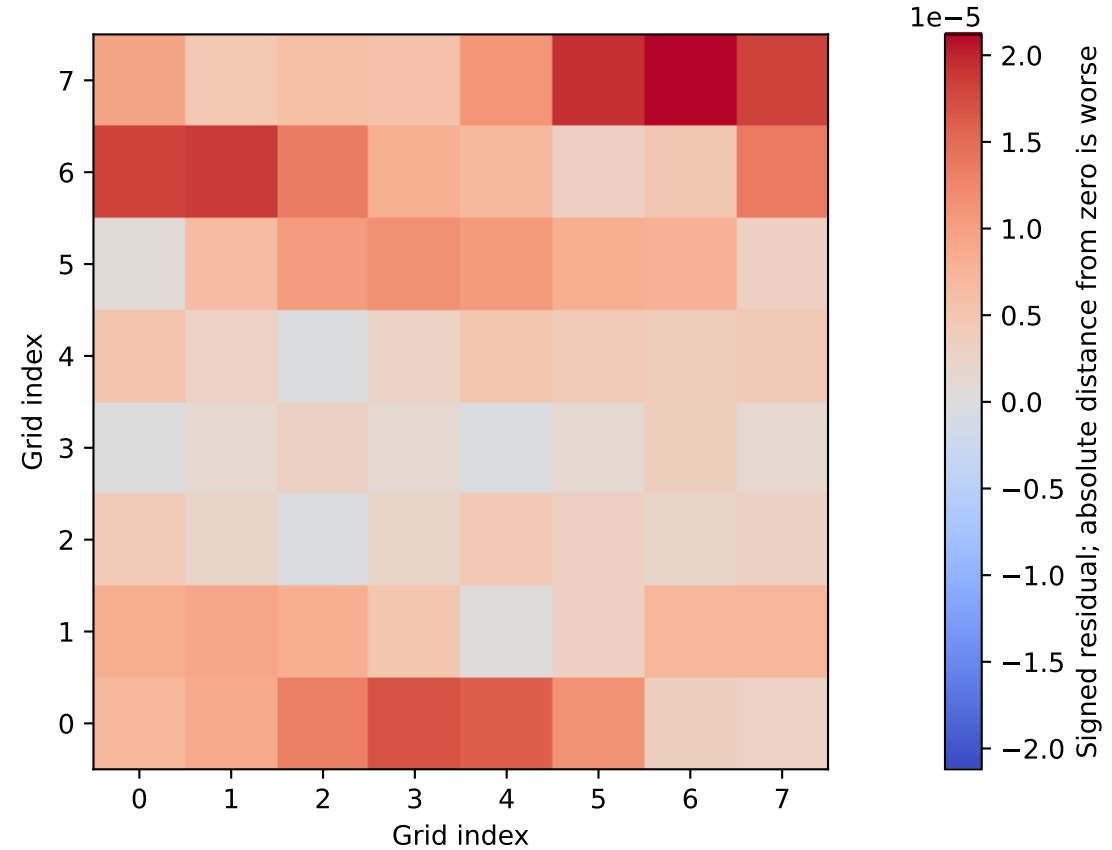
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## D01 Spatial residual Theirs: df [continuum]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

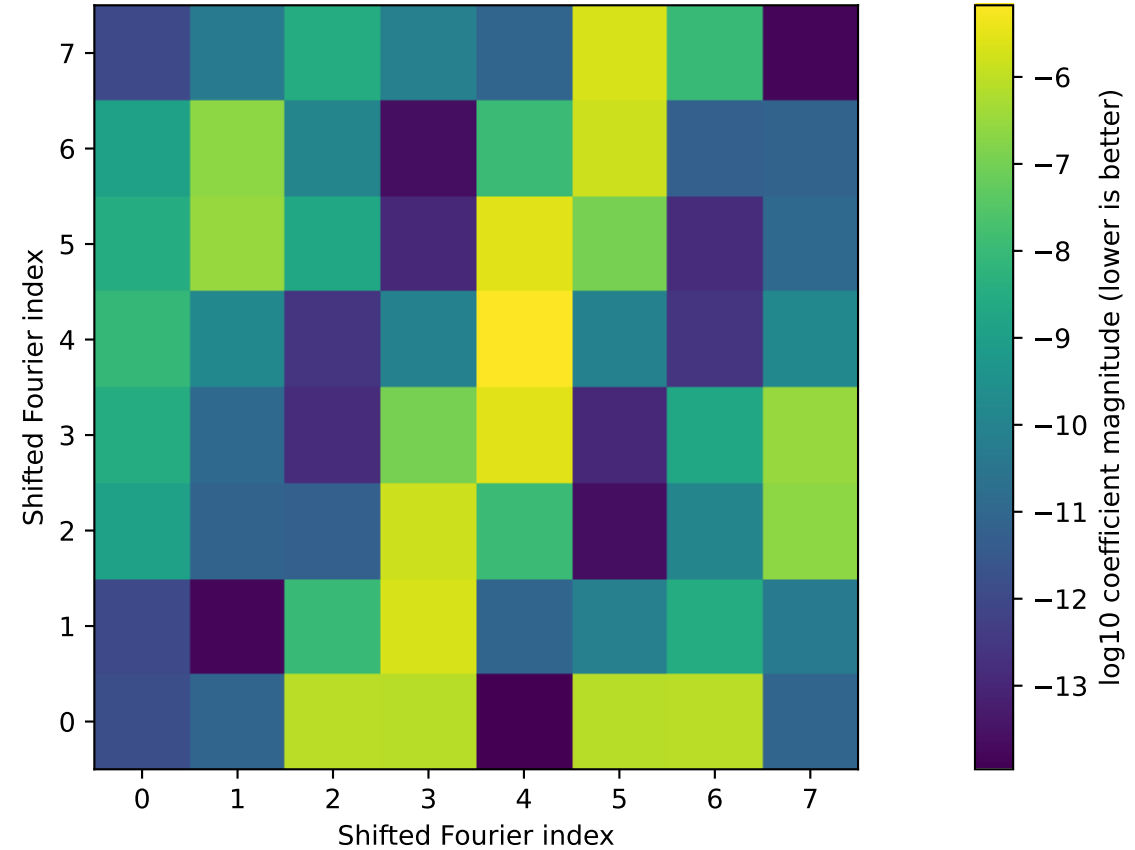
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Theirs: df [continuum]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

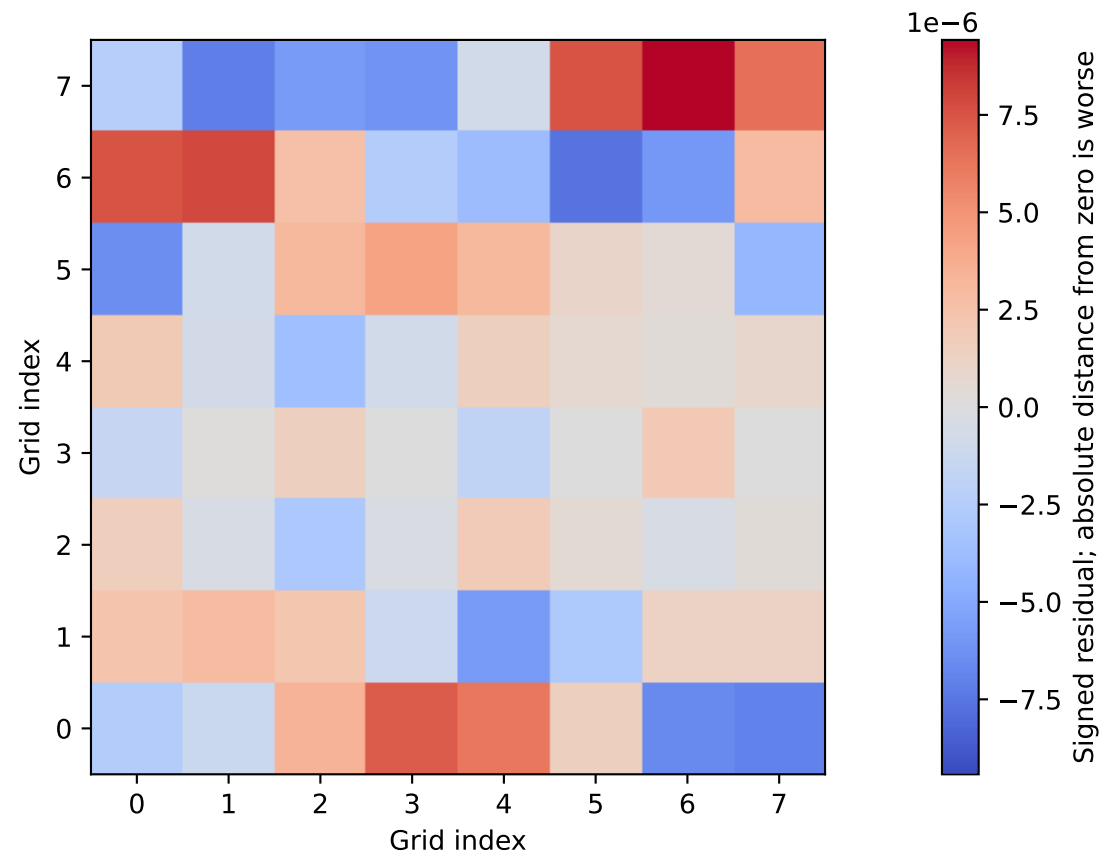
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## D01 Spatial residual Analytic control: quad2\_fixed [continuum]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

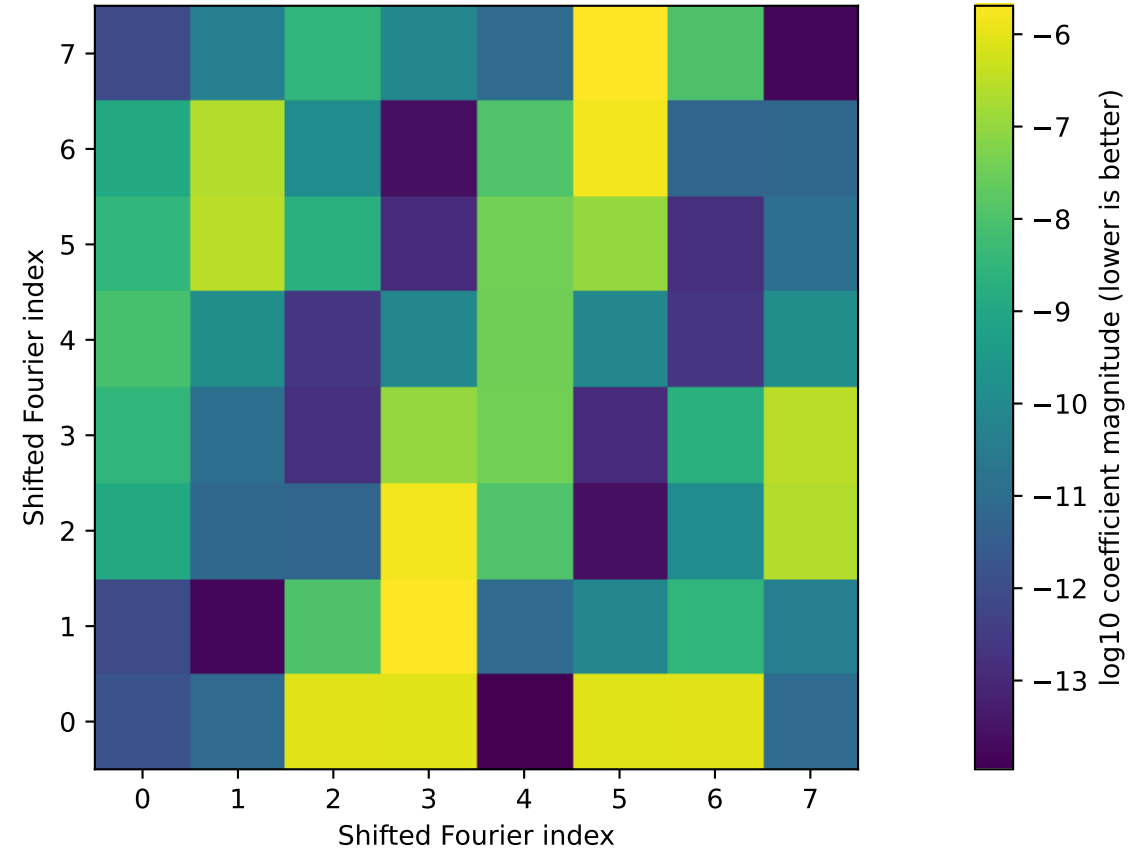
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: quad2\_fixed [continuum]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

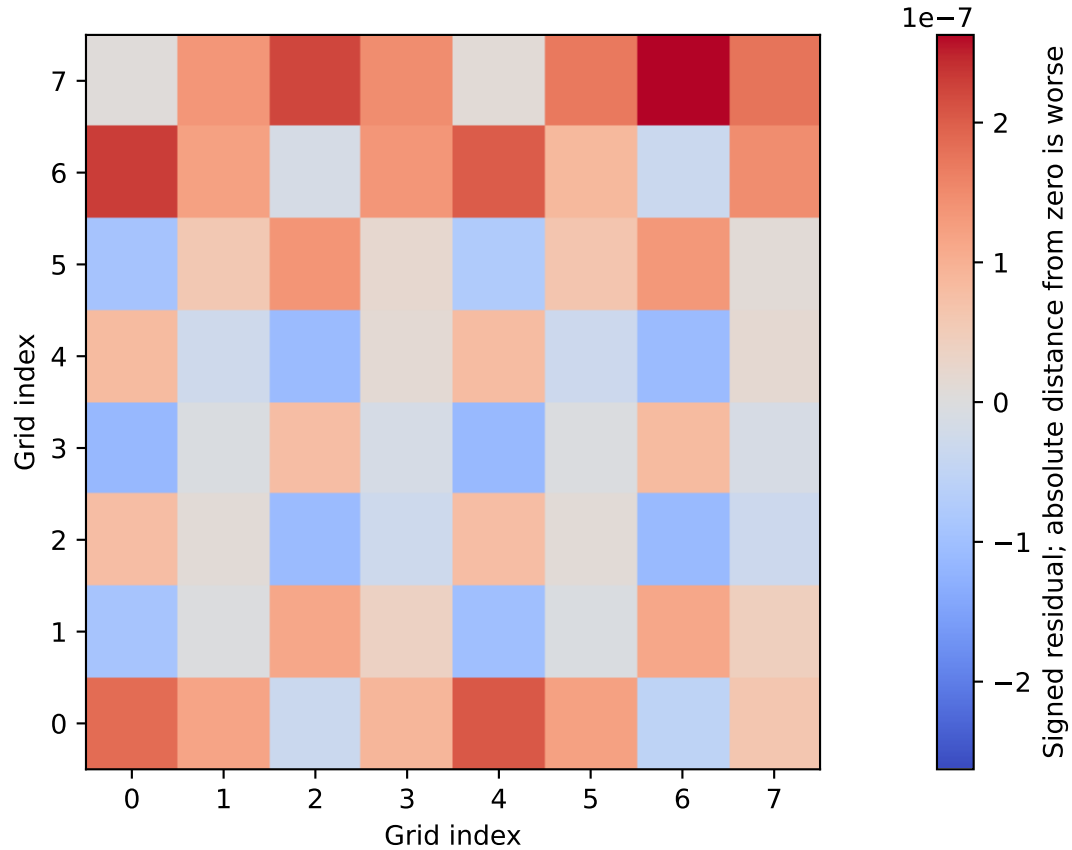
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

D01 Spatial residual Analytic control: analytic\_quad\_cubic [continuum]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

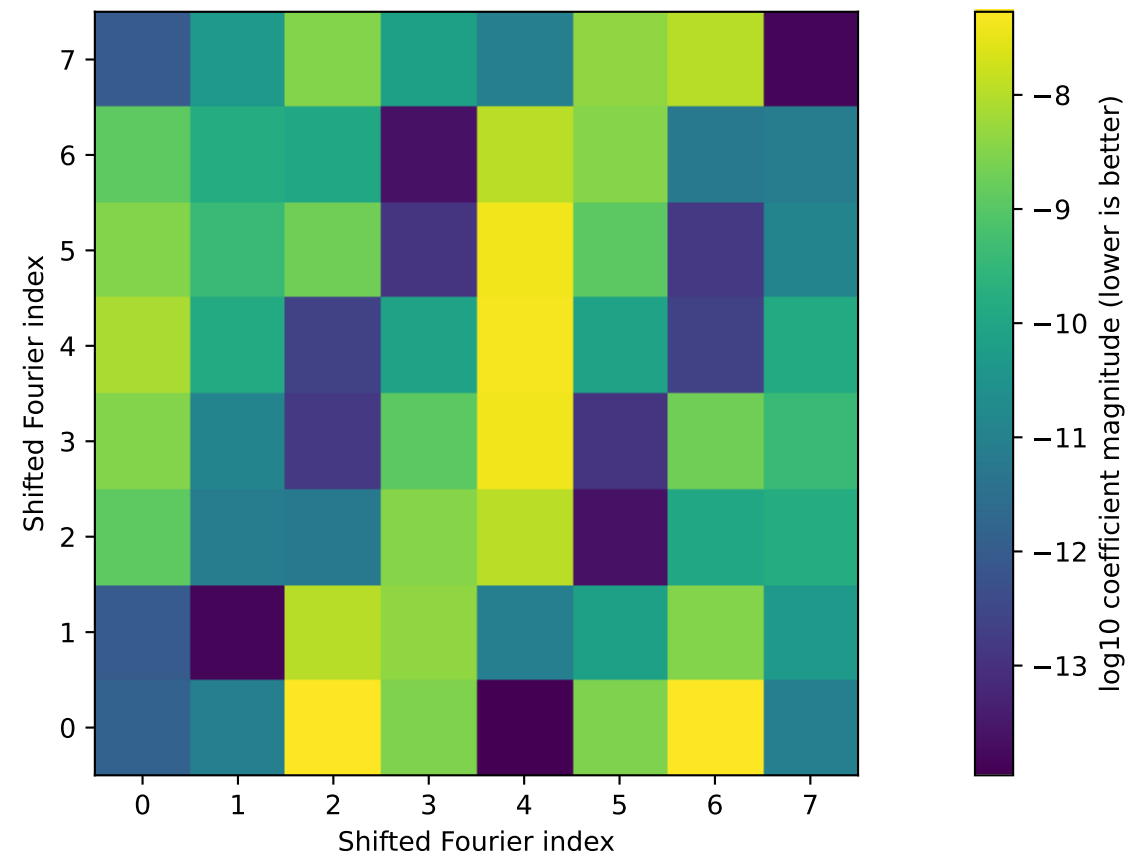
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: analytic\_quad\_cubic [continuum]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

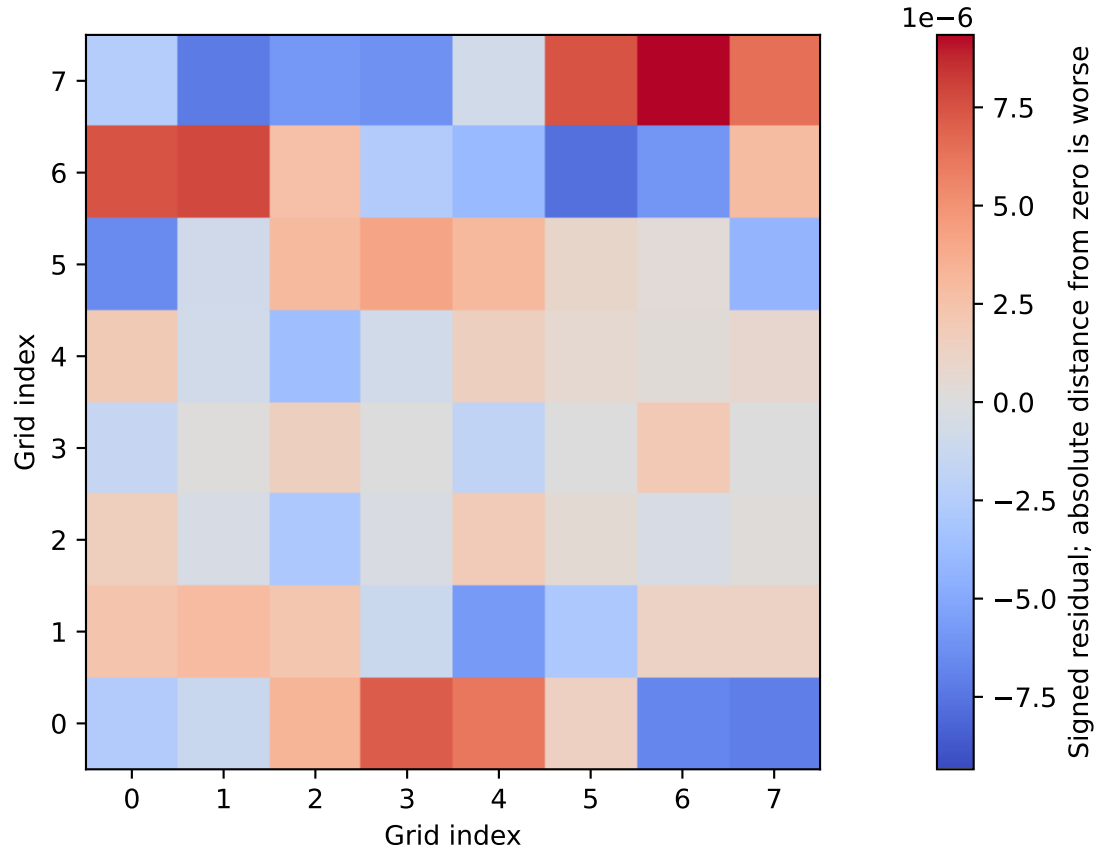
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

D01 Spatial residual Analytic control: scalar\_bounded\_oracle [continuum]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

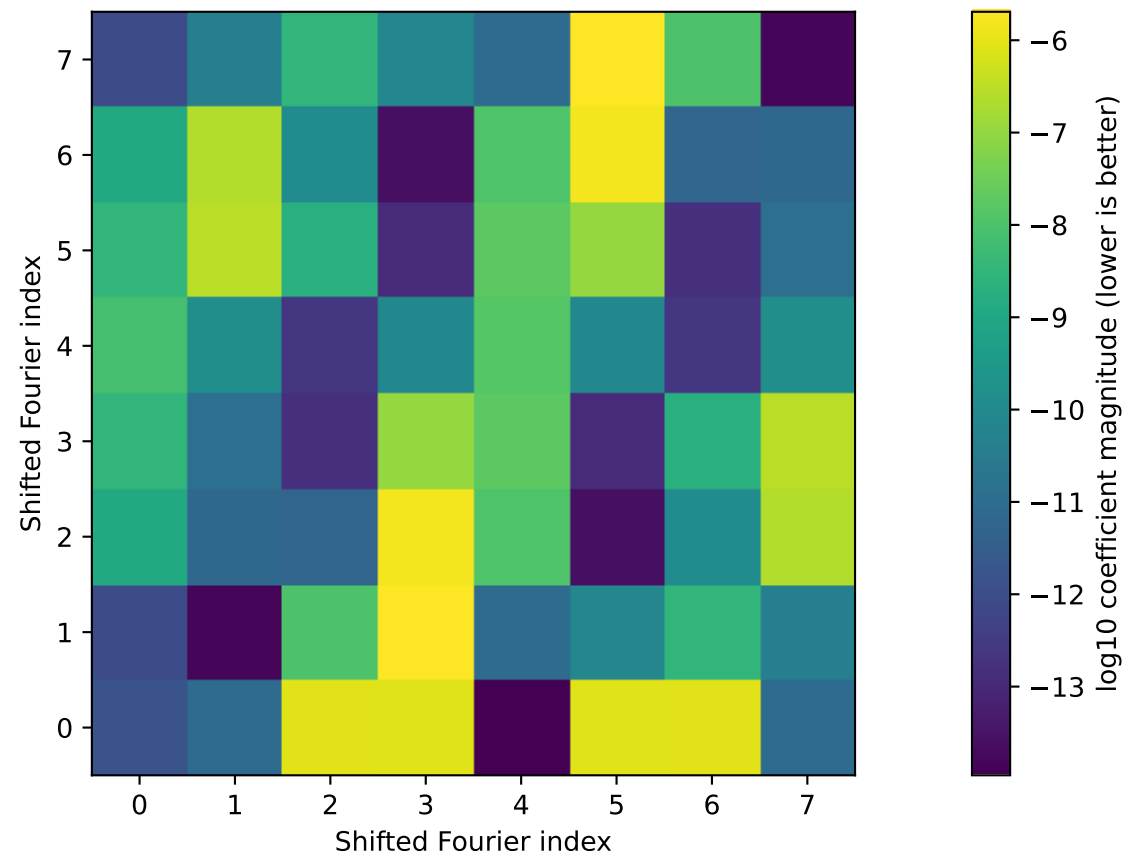
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: scalar\_bounded\_oracle [continuum]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

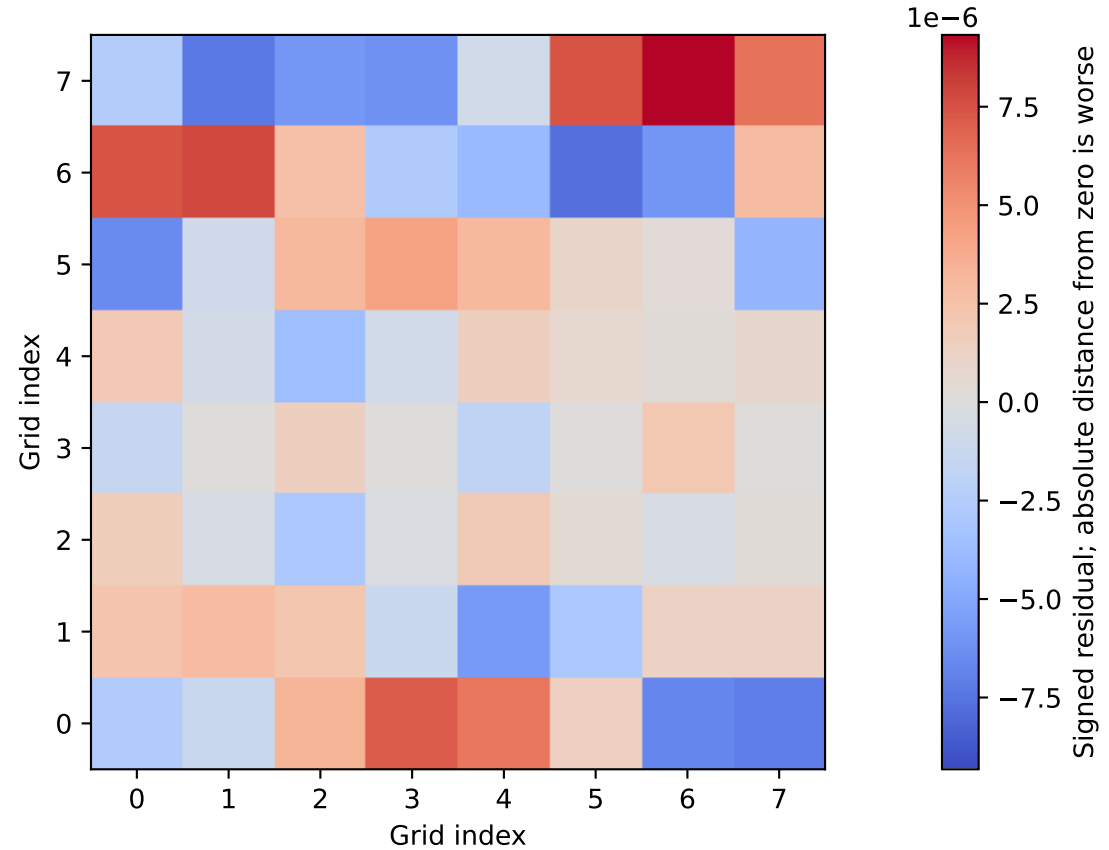
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spatial residual Analytic control: interaction\_channel\_bounded\_oracle [continuum]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

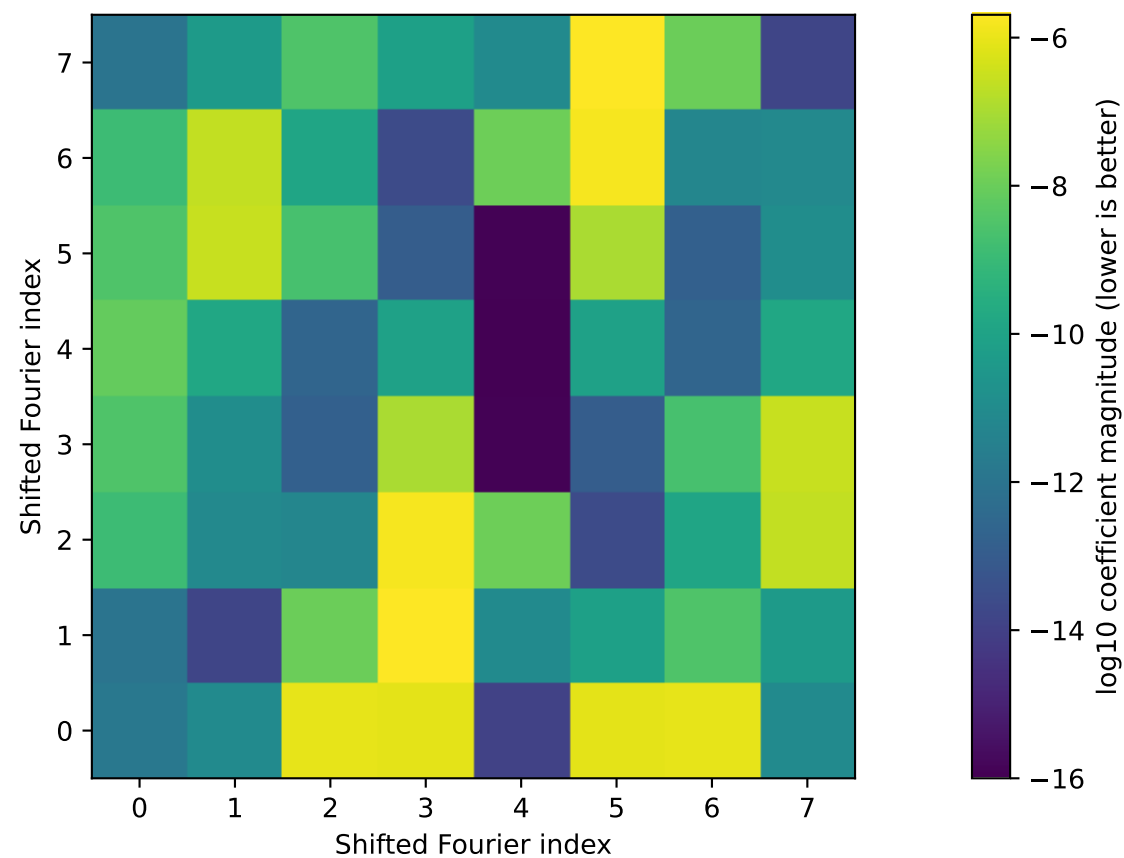
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: interaction\_channel\_bounded\_oracle [continuum]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

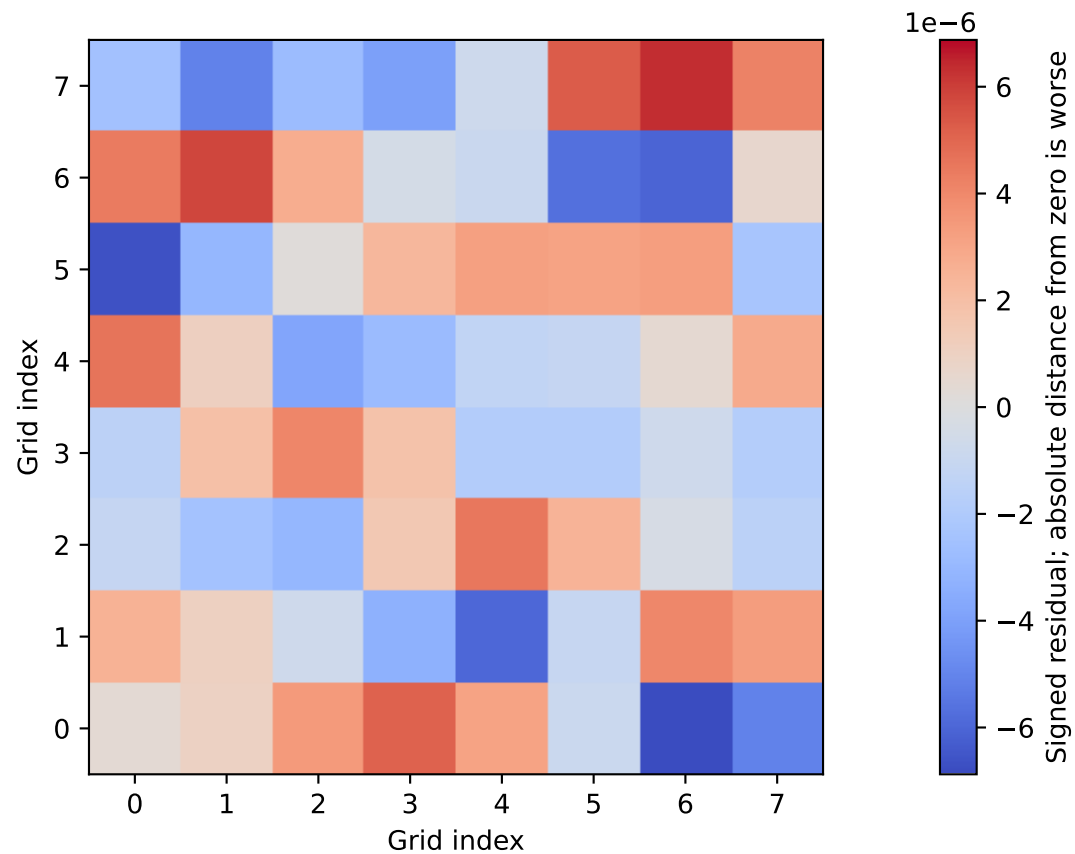
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## D01 Spatial residual Analytic control: quadratic\_cubic\_bounded\_oracle [continuum]



READ: White/zero is better; larger absolute red or blue residual is worse. Spatial indices have no favorable direction. This is the predeclared representative high-pair parent, not a pooled comparison.

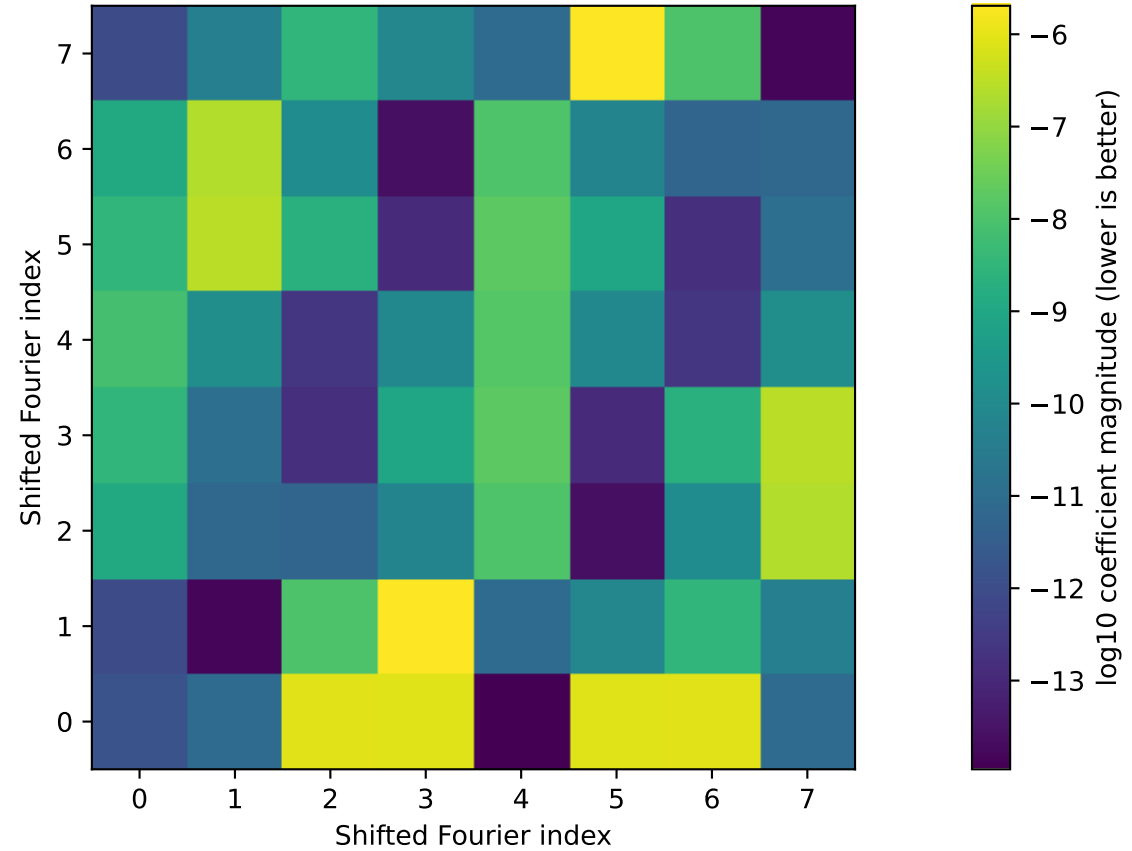
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D01 Spectral residual Analytic control: quadratic\_cubic\_bounded\_oracle [continuum]



READ: Lower color values mean smaller residual Fourier magnitude; a 1e-16 display floor is explicit. Signed complex coefficients remain in raw chart data and NPZ sources.

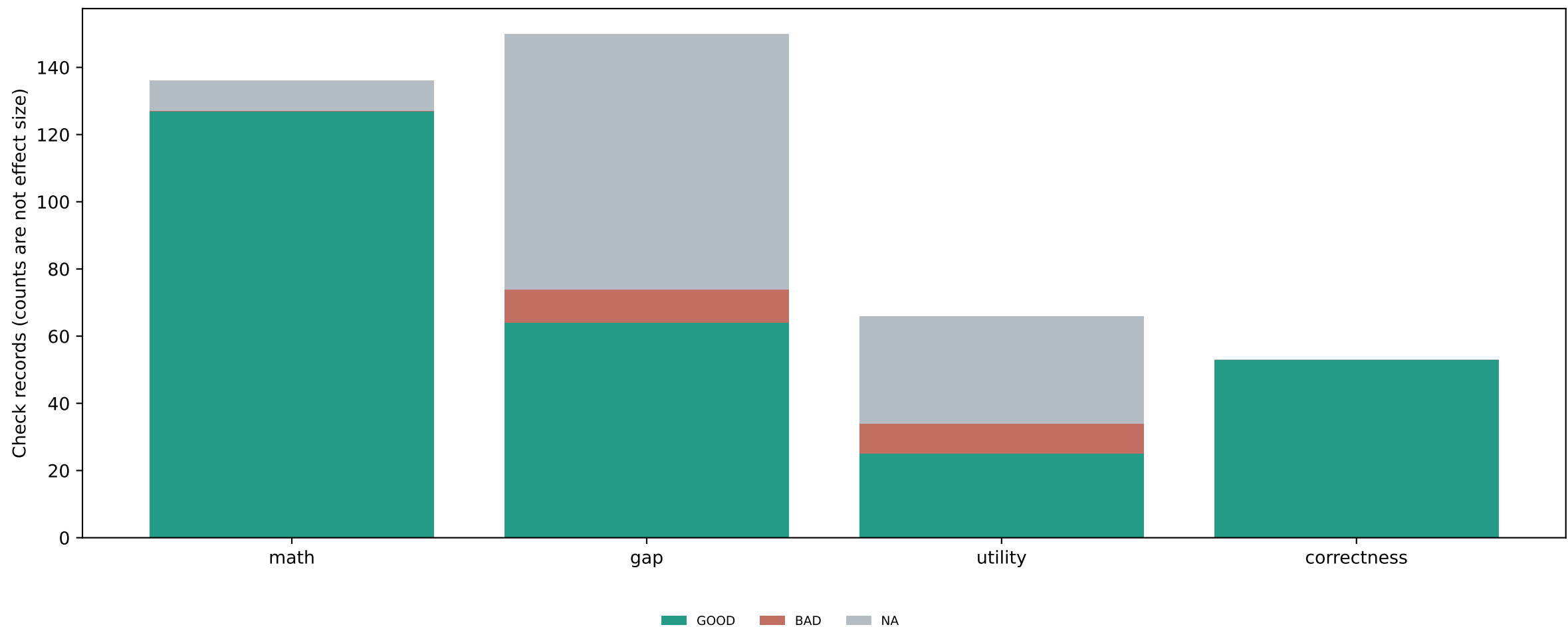
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# Mathematics gap and utility outcomes



READ: GOOD meets a declared check; BAD fails it; NA is unresolved. More passing checks do not prove a theorem, improve statistical precision or establish a model advantage.

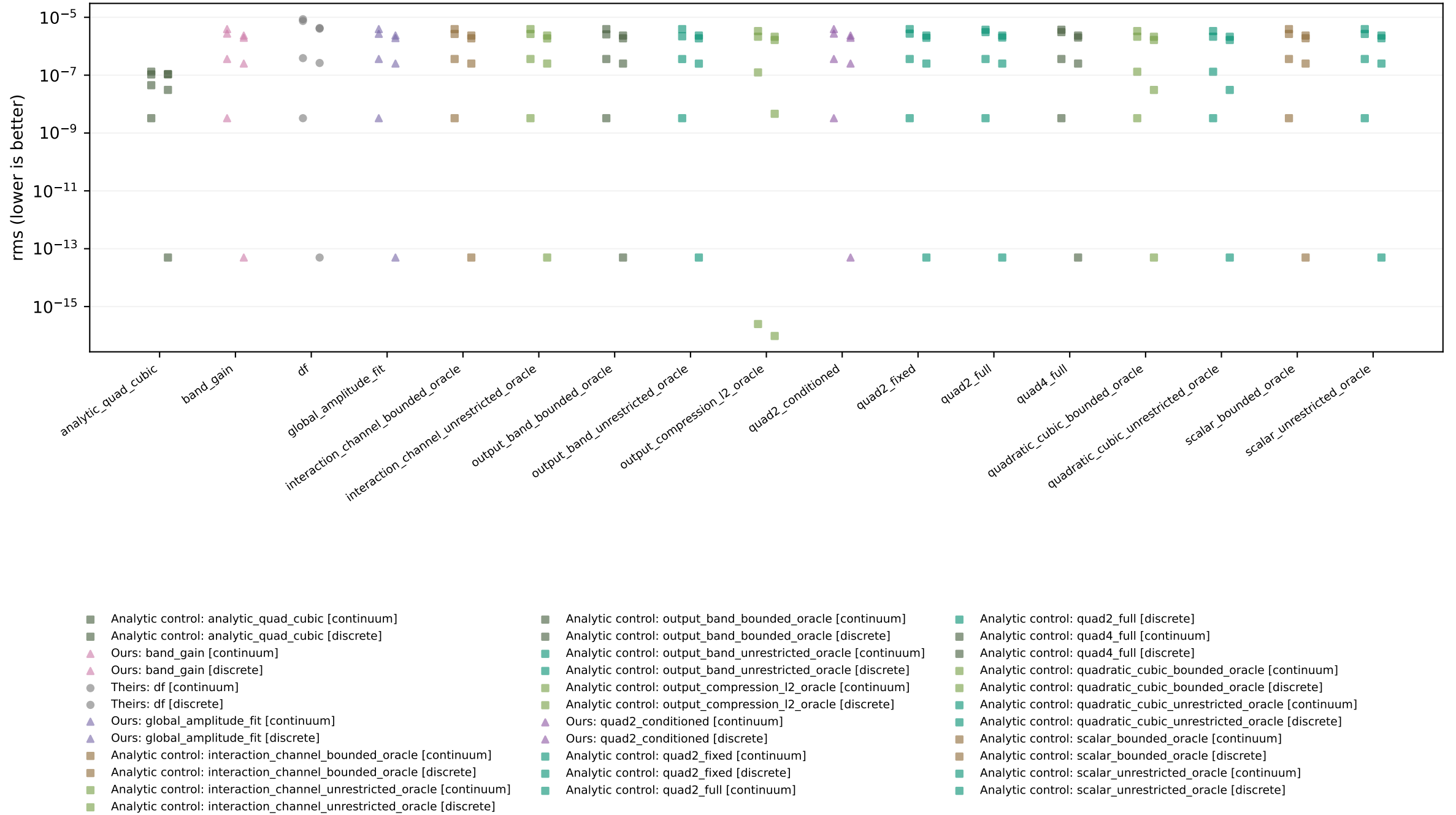
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

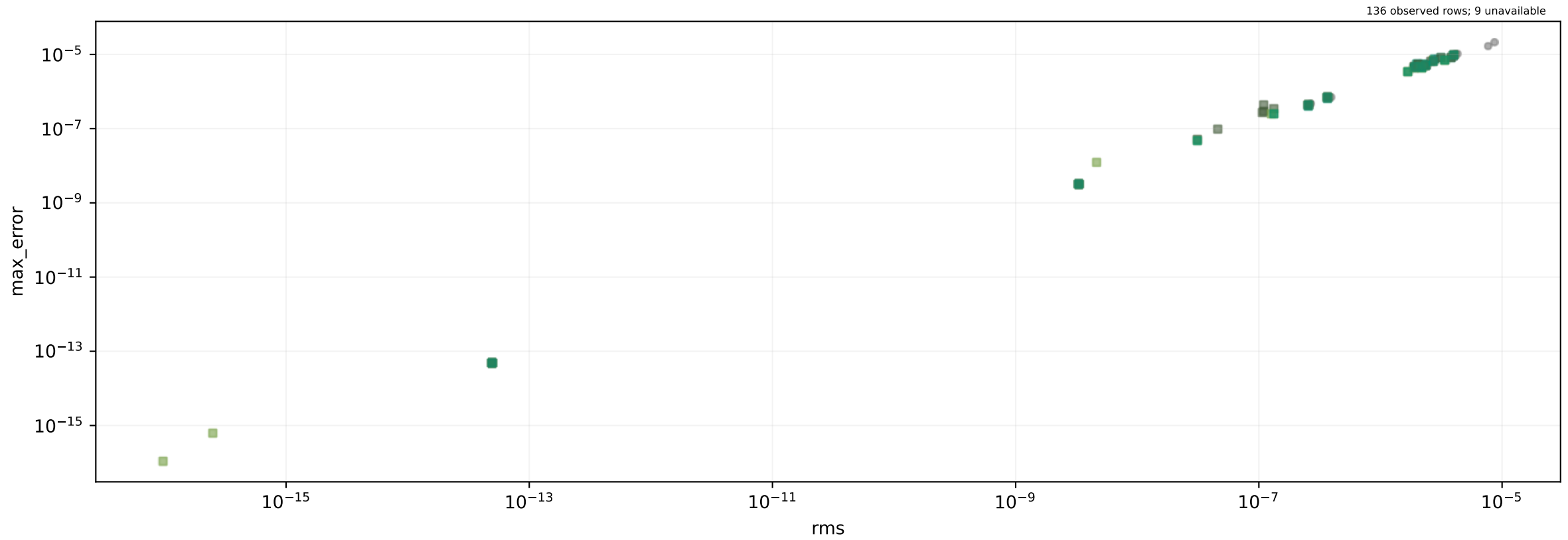
Missing evidence is NA.

# D01 Scalar and channel correction floors



READ: Y lower is a smaller correction-space error floor. X lists methods, not a ranking. Oracle fits use references and are not deployable. Values below reference uncertainty are unresolved. Compare identical field/operator/h/nodes/cutoff; offset groups distinguish tracks.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D01 Residual RMS and maximum error



- |                                                                         |                                                                 |                                                                     |
|-------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| ■ Analytic control: analytic_quad_cubic [continuum]                     | ■ Analytic control: output_band_bounded_oracle [continuum]      | ■ Analytic control: quad2_full [discrete]                           |
| ■ Analytic control: analytic_quad_cubic [discrete]                      | ■ Analytic control: output_band_bounded_oracle [discrete]       | ■ Analytic control: quad4_full [continuum]                          |
| ▲ Ours: band_gain [continuum]                                           | ■ Analytic control: output_band_unrestricted_oracle [continuum] | ■ Analytic control: quad4_full [discrete]                           |
| ▲ Ours: band_gain [discrete]                                            | ■ Analytic control: output_band_unrestricted_oracle [discrete]  | ■ Analytic control: quadratic_cubic_bounded_oracle [continuum]      |
| ● Theirs: df [continuum]                                                | ■ Analytic control: output_compression_l2_oracle [continuum]    | ■ Analytic control: quadratic_cubic_bounded_oracle [discrete]       |
| ● Theirs: df [discrete]                                                 | ■ Analytic control: output_compression_l2_oracle [discrete]     | ■ Analytic control: quadratic_cubic_unrestricted_oracle [continuum] |
| ▲ Ours: global_amplitude_fit [continuum]                                | ▲ Ours: quad2_conditioned [continuum]                           | ■ Analytic control: quadratic_cubic_unrestricted_oracle [discrete]  |
| ▲ Ours: global_amplitude_fit [discrete]                                 | ▲ Ours: quad2_conditioned [discrete]                            | ■ Analytic control: scalar_bounded_oracle [continuum]               |
| ■ Analytic control: interaction_channel_bounded_oracle [continuum]      | ■ Analytic control: quad2_fixed [continuum]                     | ■ Analytic control: scalar_bounded_oracle [discrete]                |
| ■ Analytic control: interaction_channel_bounded_oracle [discrete]       | ■ Analytic control: quad2_fixed [discrete]                      | ■ Analytic control: scalar_unrestricted_oracle [continuum]          |
| ■ Analytic control: interaction_channel_unrestricted_oracle [continuum] | ■ Analytic control: quad2_full [continuum]                      | ■ Analytic control: scalar_unrestricted_oracle [discrete]           |
| ■ Analytic control: interaction_channel_unrestricted_oracle [discrete]  |                                                                 |                                                                     |

READ: Lower on each axis is better. A small RMS does not imply small worst-cell error. Squared-L2 projection is not a minimax or composite-loss optimum; compare the same parent, equation, h, nodes and cutoff.

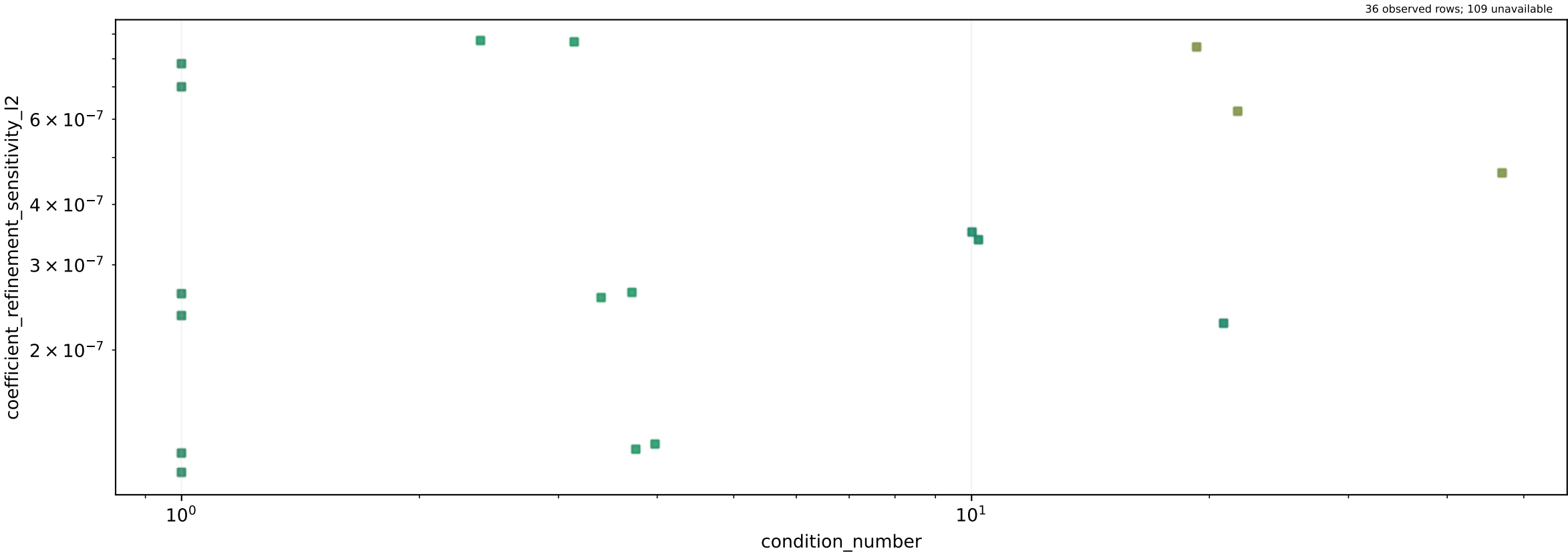
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

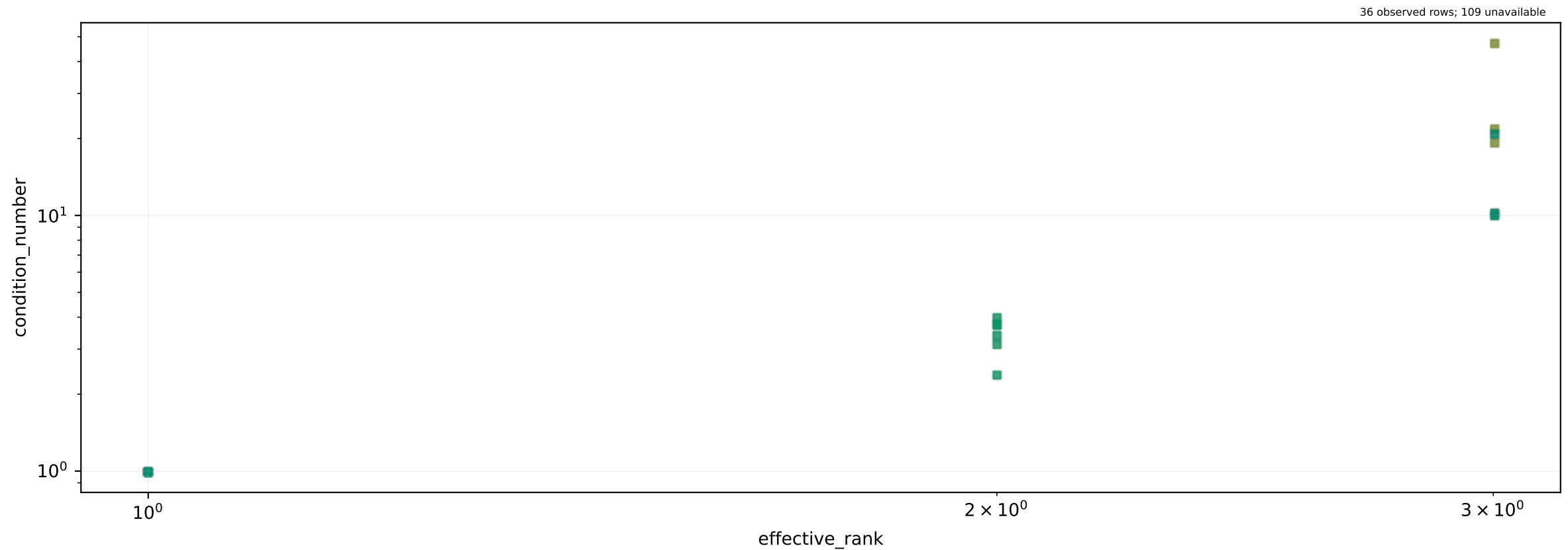
Missing evidence is NA.

# D01 Coefficient sensitivity



READ: Lower Y means fitted coefficients change less under reference refinement. Ill-conditioning can make gains unidentifiable even with a small residual.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D01 Basis conditioning and effective rank



- |                                                                       |                                                                   |                                                                  |
|-----------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------|
| Analytic control: interaction_channel_bounded_oracle [continuum]      | Analytic control: output_band_unrestricted_oracle [continuum]     | Analytic control: quadratic_cubic_unrestricted_oracle [discrete] |
| Analytic control: interaction_channel_bounded_oracle [discrete]       | Analytic control: output_band_unrestricted_oracle [discrete]      | Analytic control: scalar_bounded_oracle [continuum]              |
| Analytic control: interaction_channel_unrestricted_oracle [continuum] | Analytic control: quadratic_cubic_bounded_oracle [continuum]      | Analytic control: scalar_bounded_oracle [discrete]               |
| Analytic control: interaction_channel_unrestricted_oracle [discrete]  | Analytic control: quadratic_cubic_bounded_oracle [discrete]       | Analytic control: scalar_unrestricted_oracle [continuum]         |
| Analytic control: output_band_bounded_oracle [continuum]              | Analytic control: quadratic_cubic_unrestricted_oracle [continuum] | Analytic control: scalar_unrestricted_oracle [discrete]          |
| Analytic control: output_band_bounded_oracle [discrete]               |                                                                   |                                                                  |

READ: Rank counts reference-resolved numerical directions at the recorded uncertainty threshold, not exact algebraic or theorem rank. Lower condition number is better conditioning; rank alone is not solver quality.

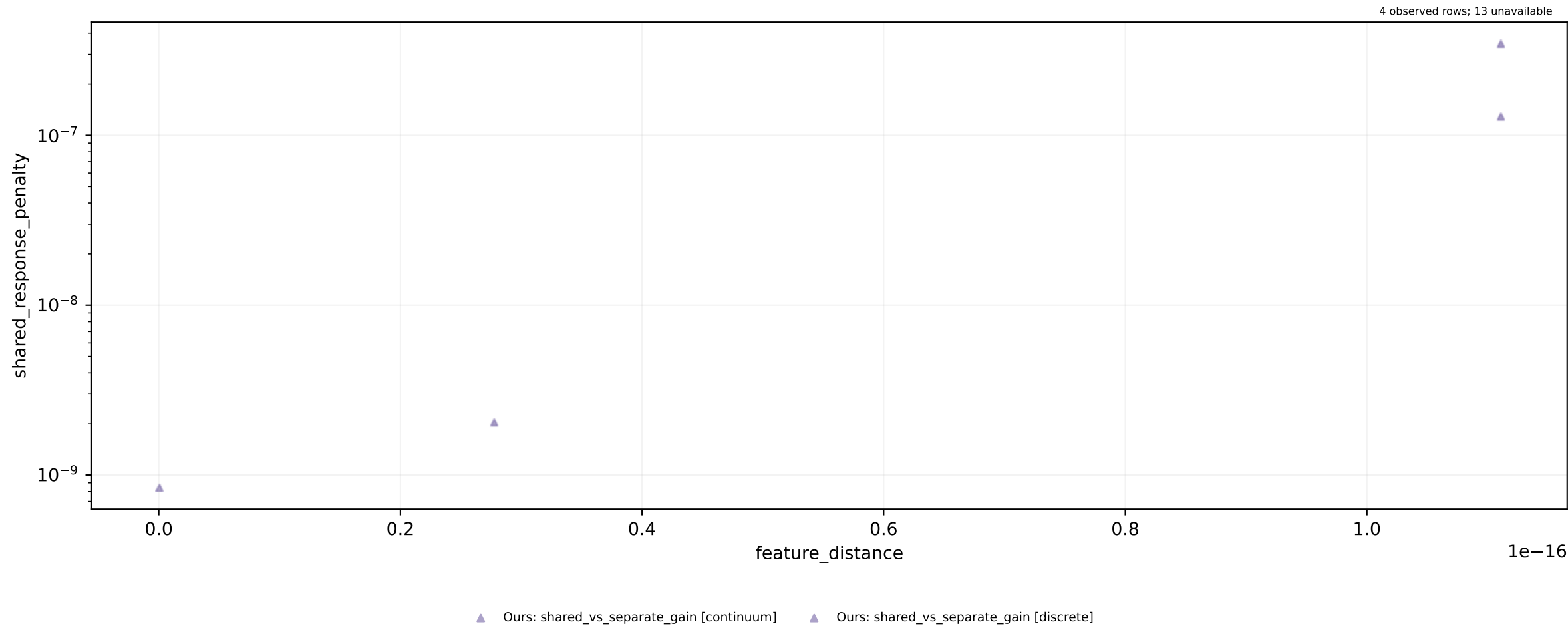
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

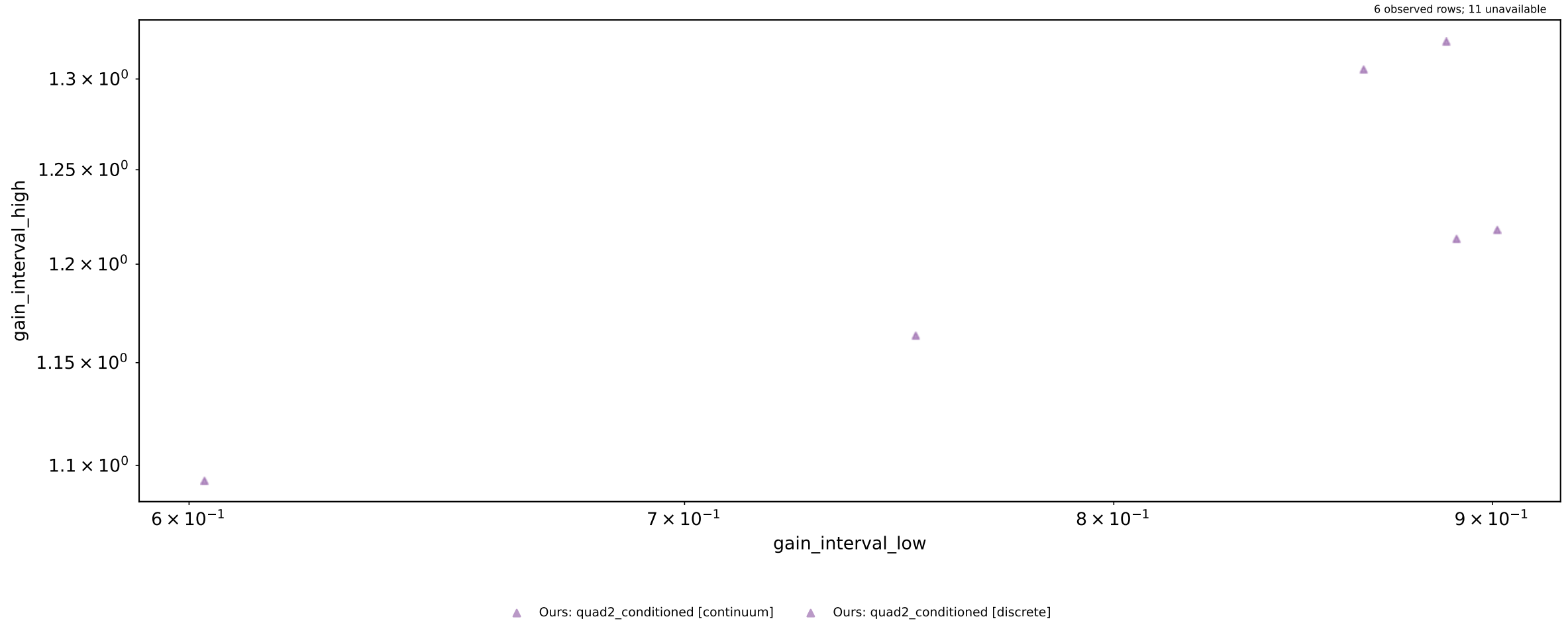
Missing evidence is NA.

D02 Feature collisions and response ambiguity



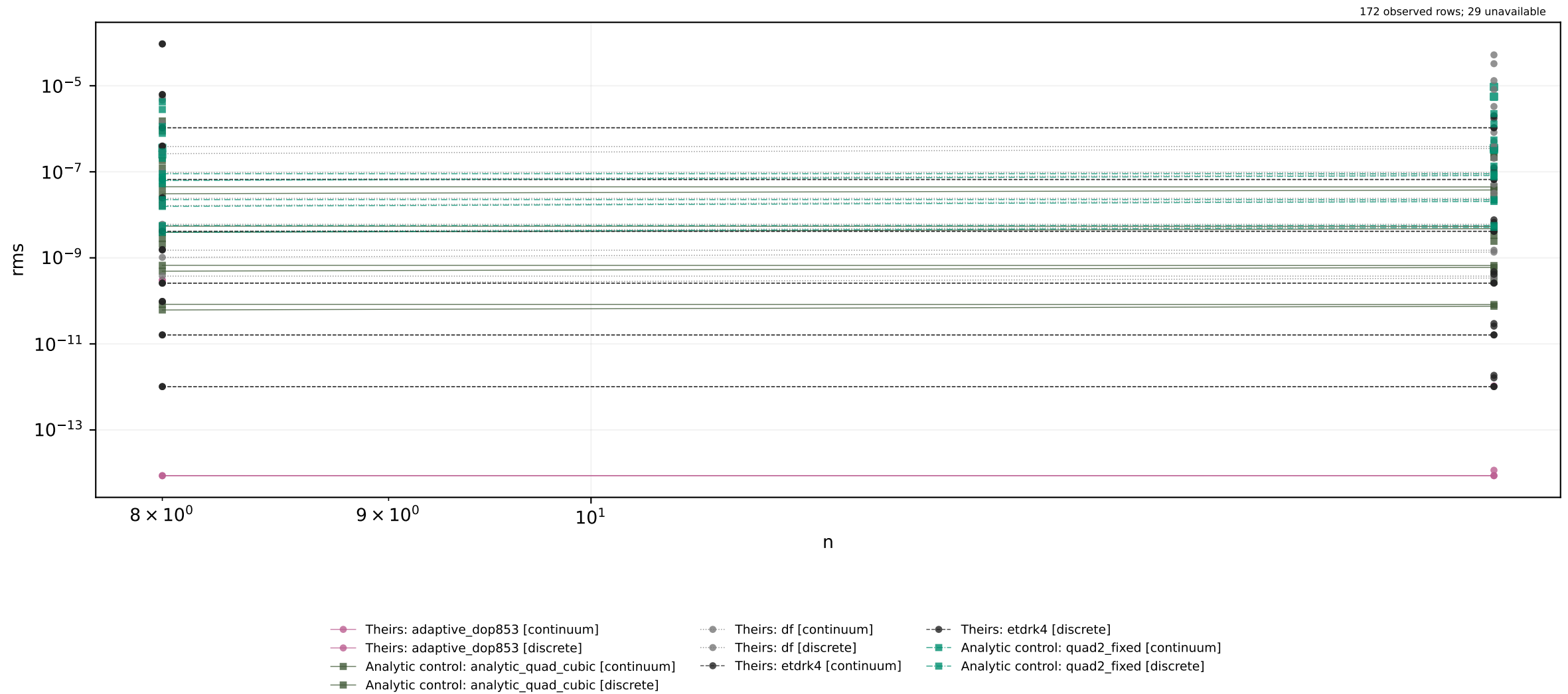
READ: A small X with materially positive Y supports insufficient conditioning information only beyond reference uncertainty and flat-response sensitivity.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

## D02 Acceptable gain intervals



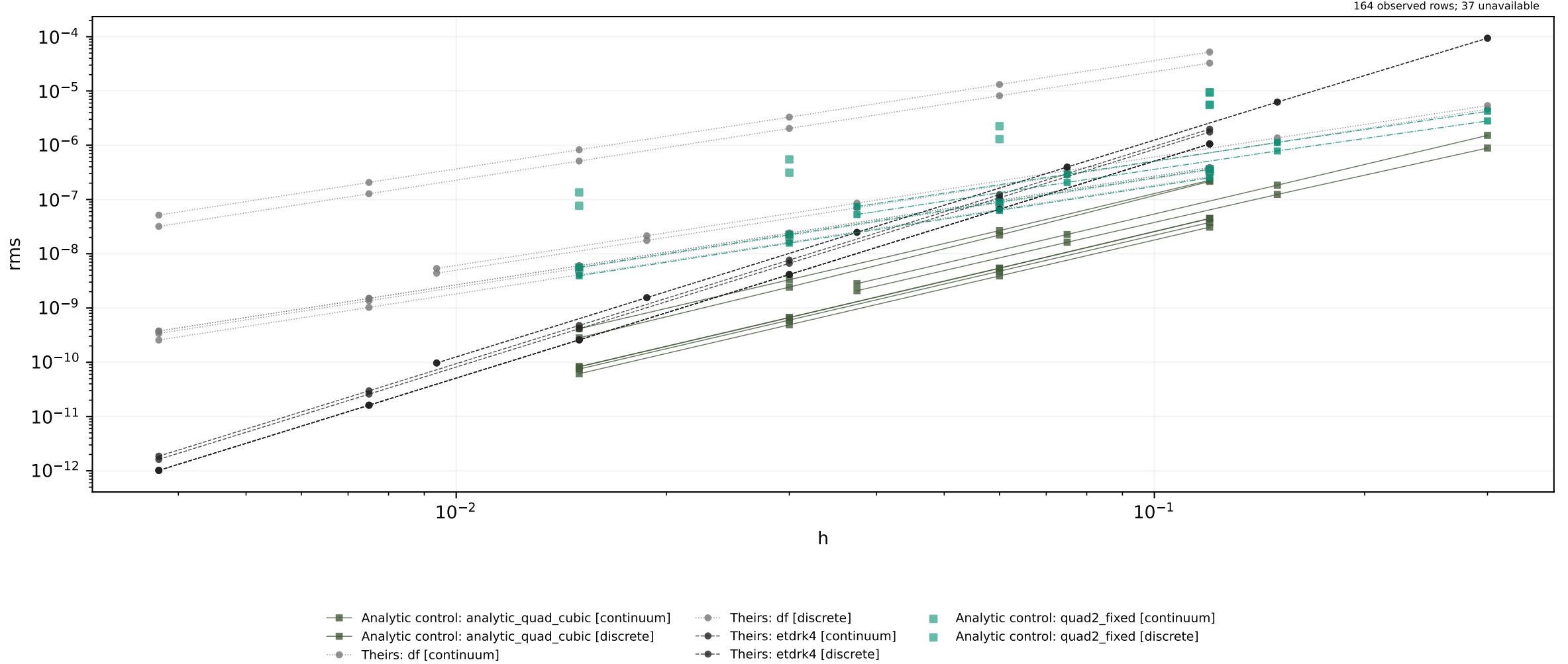
READ: Intervals are acceptable objective ranges, not confidence intervals. Separated optima alone are insufficient; interval incompatibility and excess error matter.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D03 Grid refinement at fixed workload



READ: Y lower is better; X finer grid costs more. Follow only paired parents with fixed physical bandwidth. Added scales change the workload.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D03 Timestep refinement



READ: Y lower is better; X smaller step generally costs more. Amplitude, grid and endpoint must stay fixed to interpret temporal order.

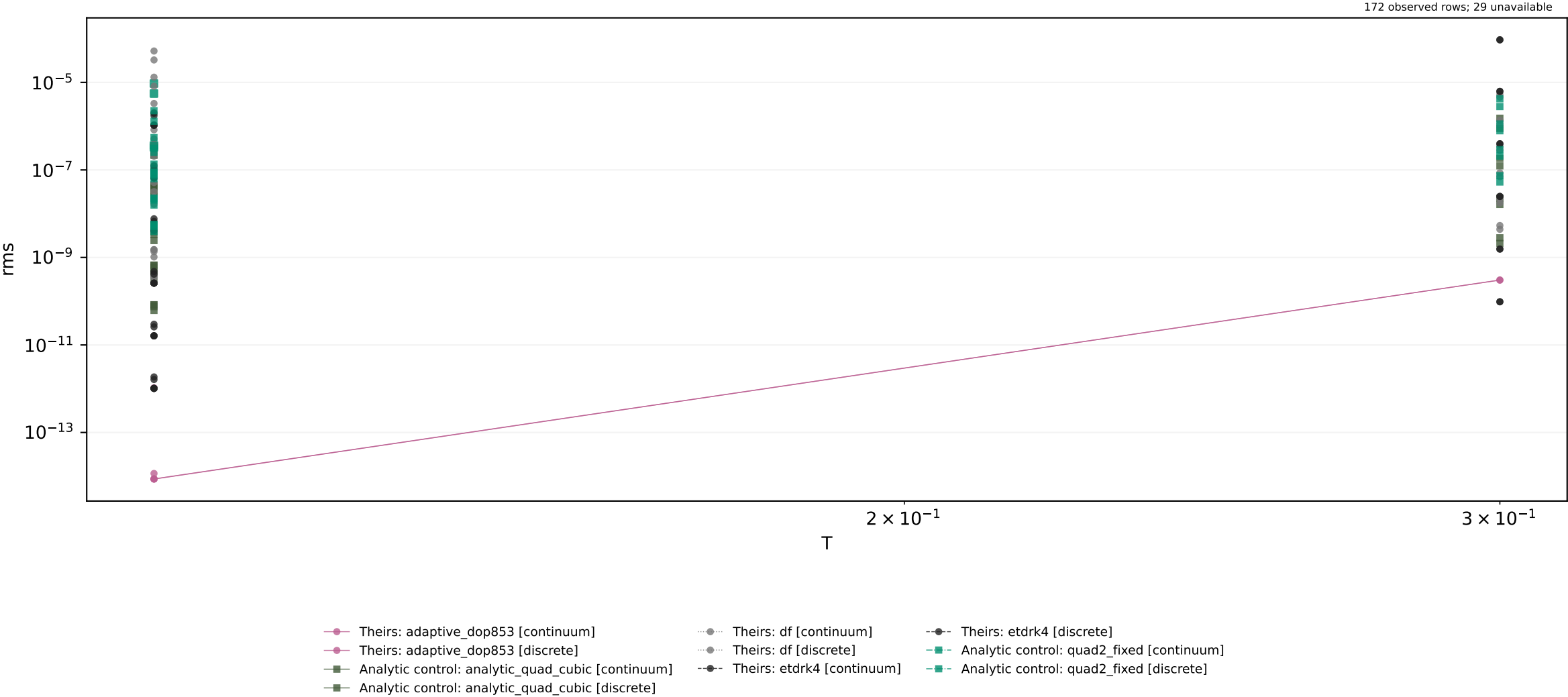
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

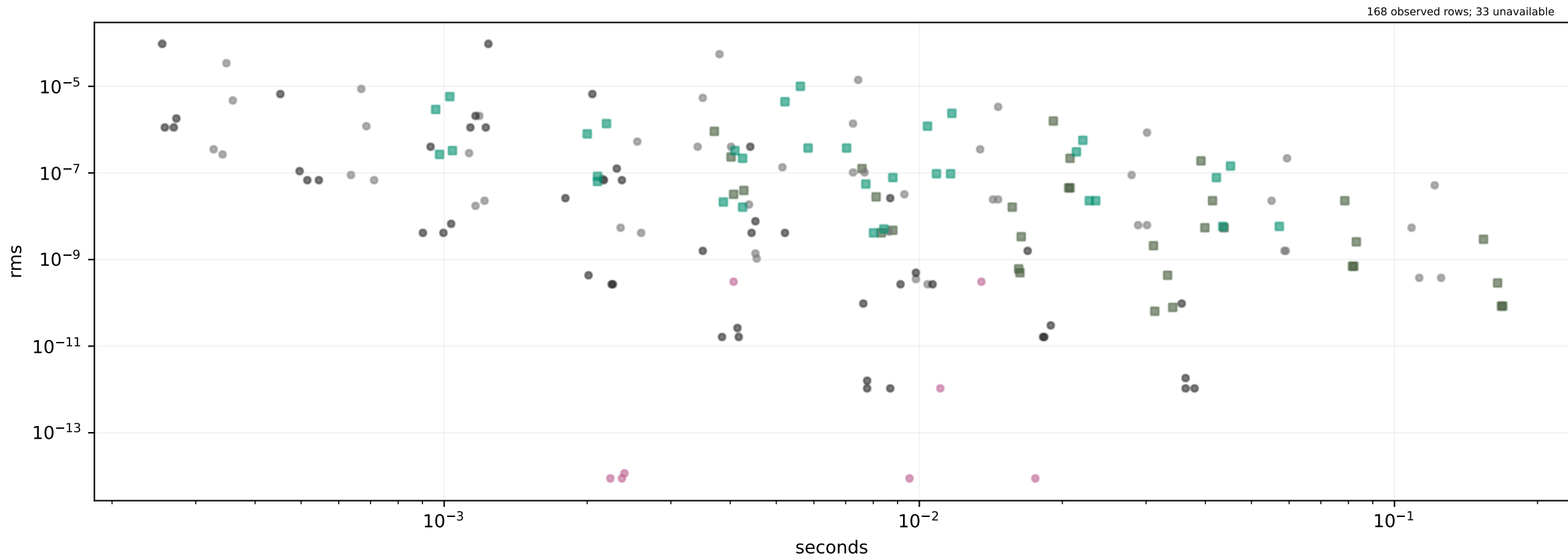
Missing evidence is NA.

# D03 Duration and complete solve error



READ: Y lower is better at the same T. A later endpoint near equilibrium may hide earlier errors. Every exact step sequence remains in raw data.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

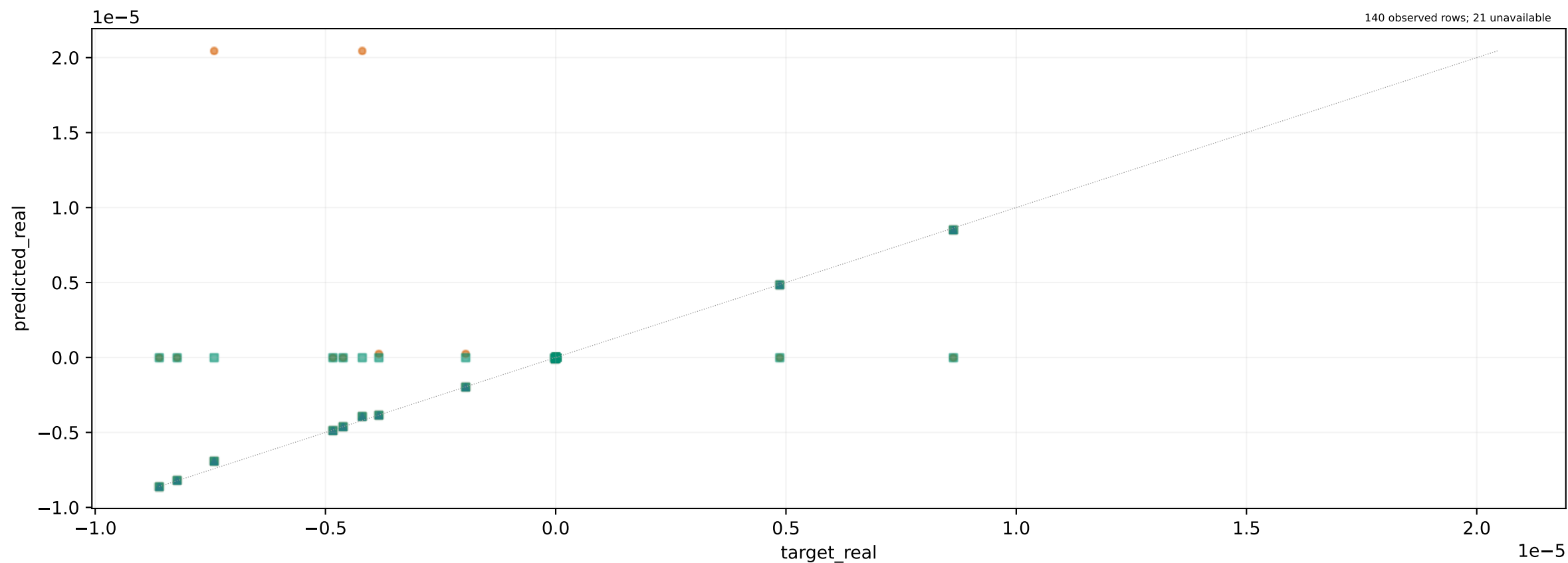
# D03 Accuracy cost observations



- Theirs: adaptive\_dop853 [continuum]
- Theirs: adaptive\_dop853 [discrete]
- Analytic control: analytic\_quad\_cubic [continuum]
- Analytic control: analytic\_quad\_cubic [discrete]
- Theirs: df [continuum]
- Theirs: df [discrete]
- Theirs: etdrk4 [discrete]
- Theirs: etdrk4 [continuum]
- Analytic control: quad2\_fixed [continuum]
- Analytic control: quad2\_fixed [discrete]

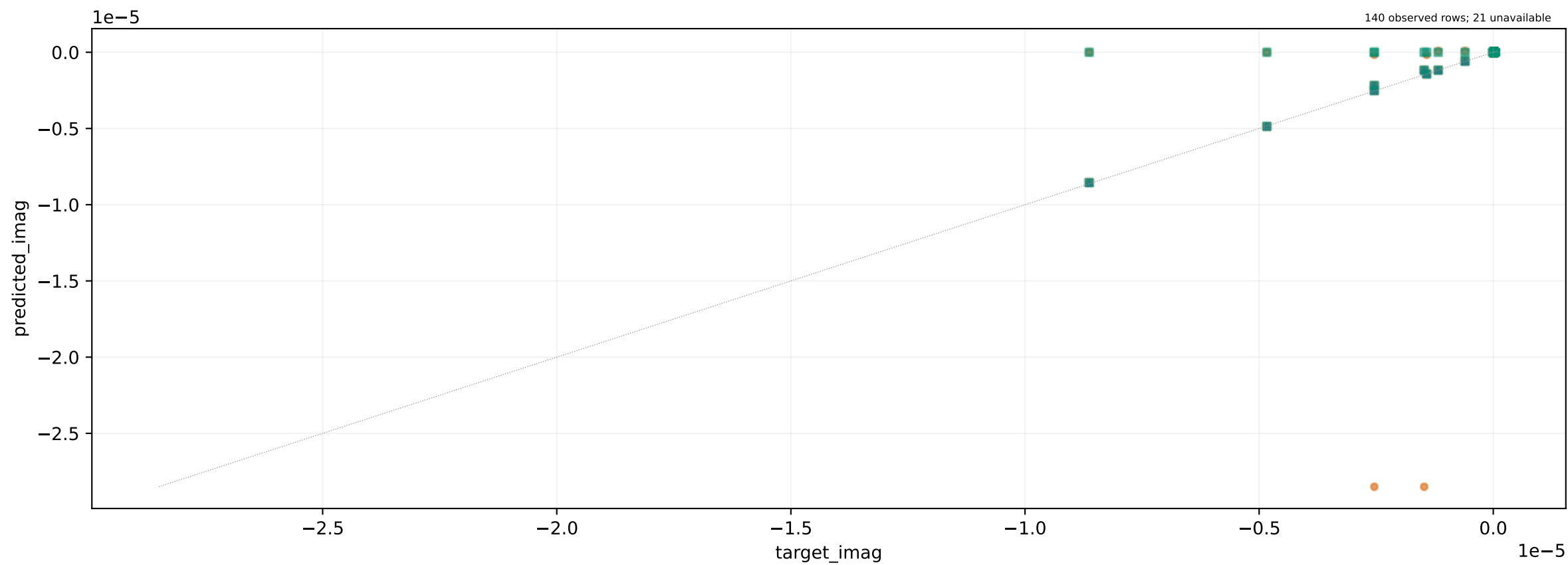
READ: Lower-left is favorable within the same track, field, grid, T and target. Unqualified latency is not time to an accurate solution.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

D04 Signed interaction coefficient error



READ: Agreement with the diagonal is better; sign matters. Neither larger X nor larger Y alone is favorable. Near-zero targets require absolute error.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

D04 Signed imaginary interaction coefficient



- ▲ Ours: band\_gain [continuum]
- ▲ Ours: band\_gain [discrete]
- Analytic control: channel\_fixed [continuum]
- Analytic control: channel\_fixed [discrete]
- ▲ Ours: channel\_neural [continuum]
- ▲ Ours: channel\_neural [discrete]
- Analytic control: quad2\_fixed [continuum]
- Analytic control: quad2\_fixed [discrete]
- Analytic control: quad2\_input [continuum]
- Analytic control: quad2\_input [discrete]
- Theirs: fno\_small [continuum]
- Theirs: fno\_small [discrete]
- ▲ Ours: quad2\_conditioned [continuum]
- ▲ Ours: quad2\_conditioned [discrete]
- ..... Exact agreement

READ: Agreement with the diagonal is better; imaginary signs carry phase. Larger magnitude alone is neither better nor worse.

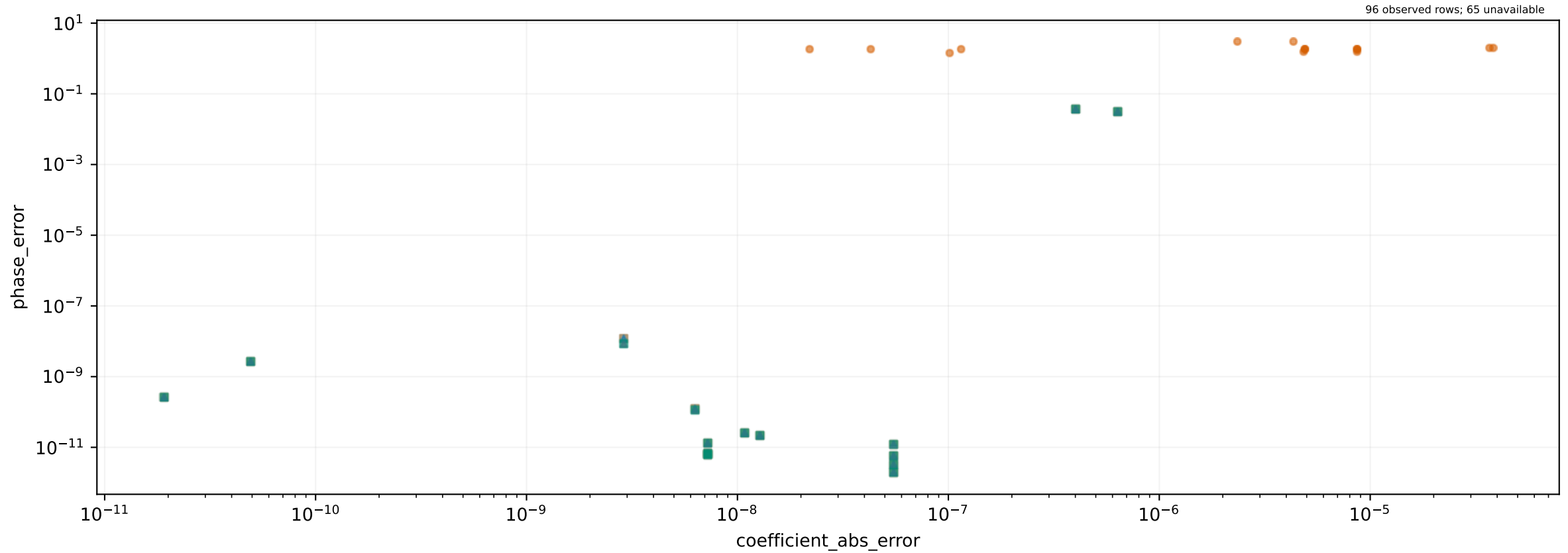
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

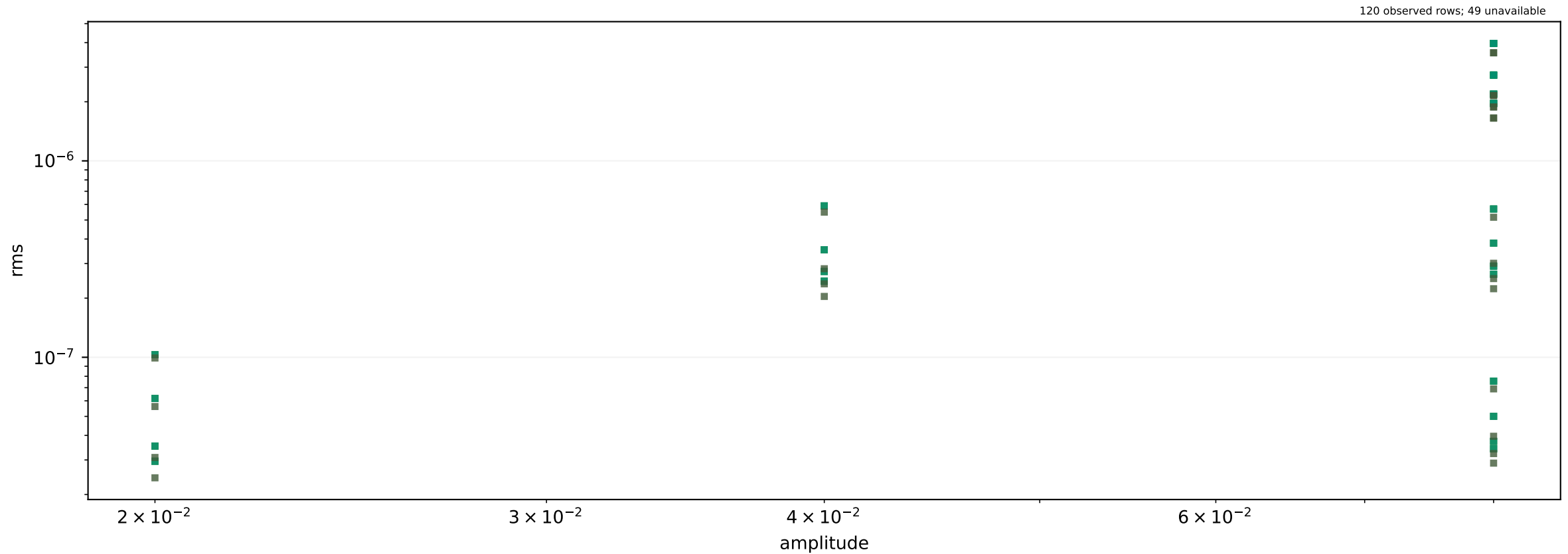
Missing evidence is NA.

# D04 Phase and coefficient magnitude error



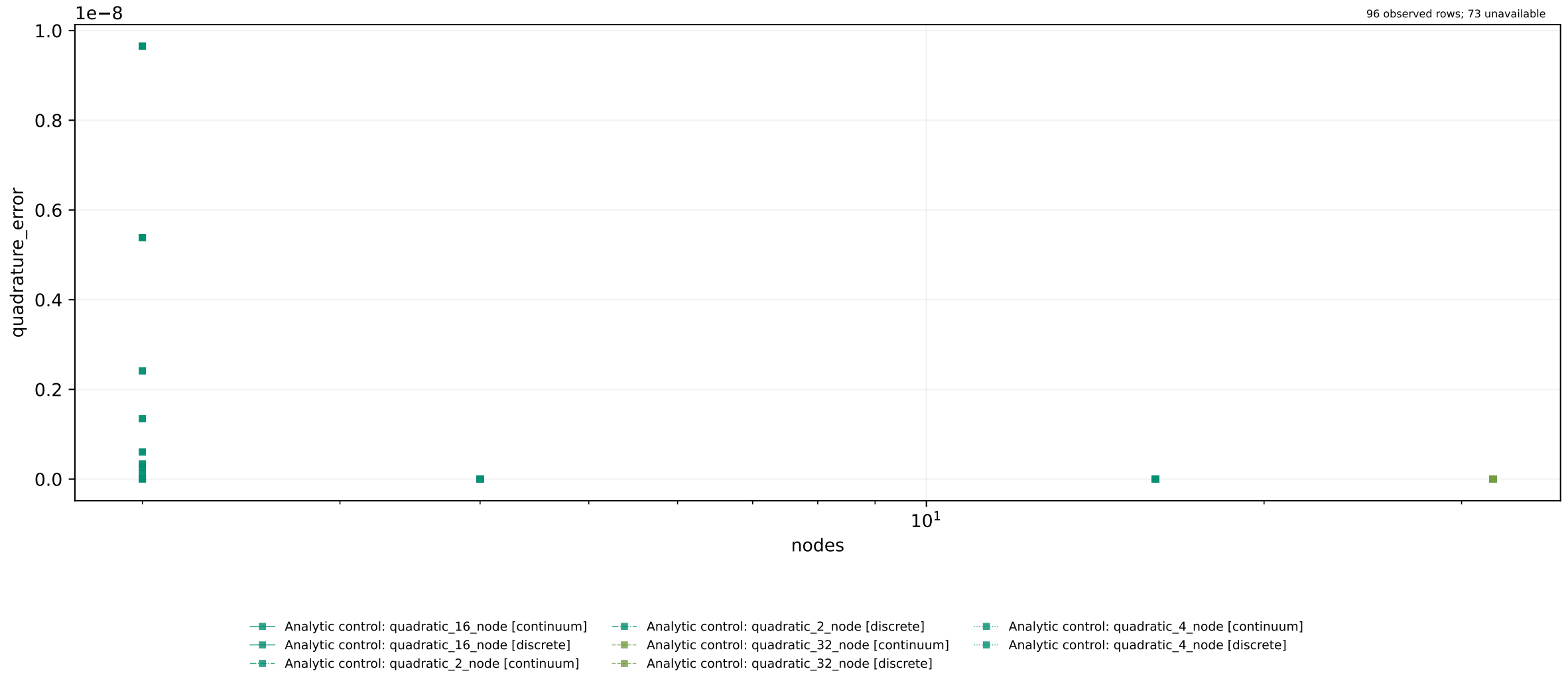
READ: Lower on both axes is better. Phase at vanishing amplitude is unresolved; nodal and dealiased products are different target equations.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D05 Fluctuation amplitude order



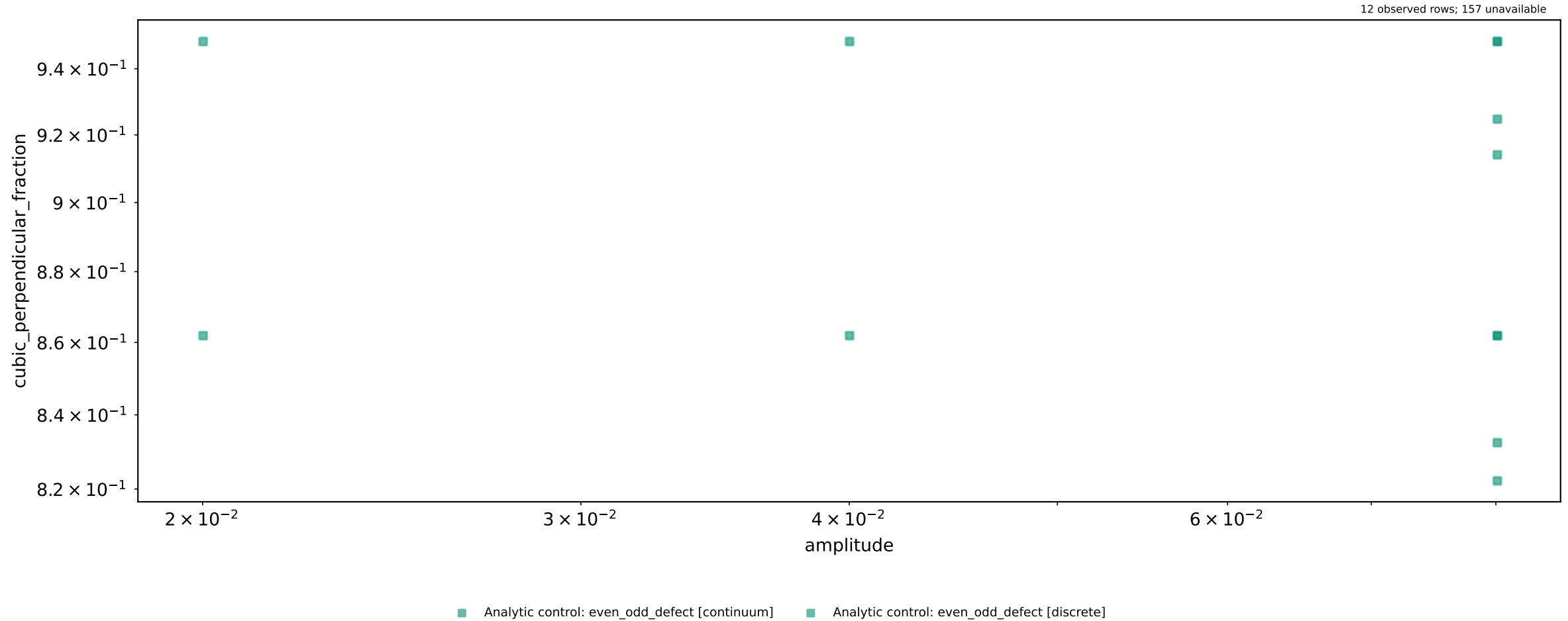
READ: Y lower is better; X varies perturbation amplitude at fixed h. Slopes are meaningful only above uncertainty and roundoff; this is not temporal order.  
 METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
 SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
 Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
 Missing evidence is NA.

## D05 Quadrature convergence



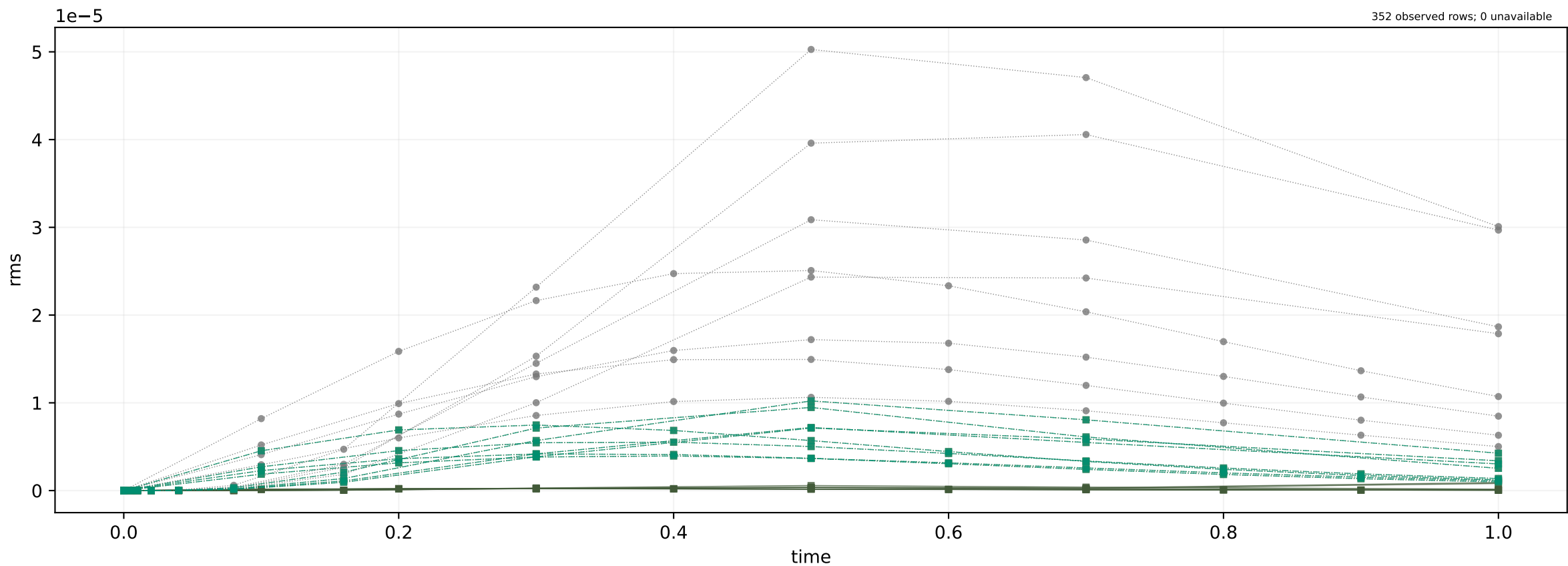
READ: Y lower is better; X more nodes adds work. Refining a quadratic integral does not introduce a missing cubic spatial direction.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D05 Cubic spatial direction



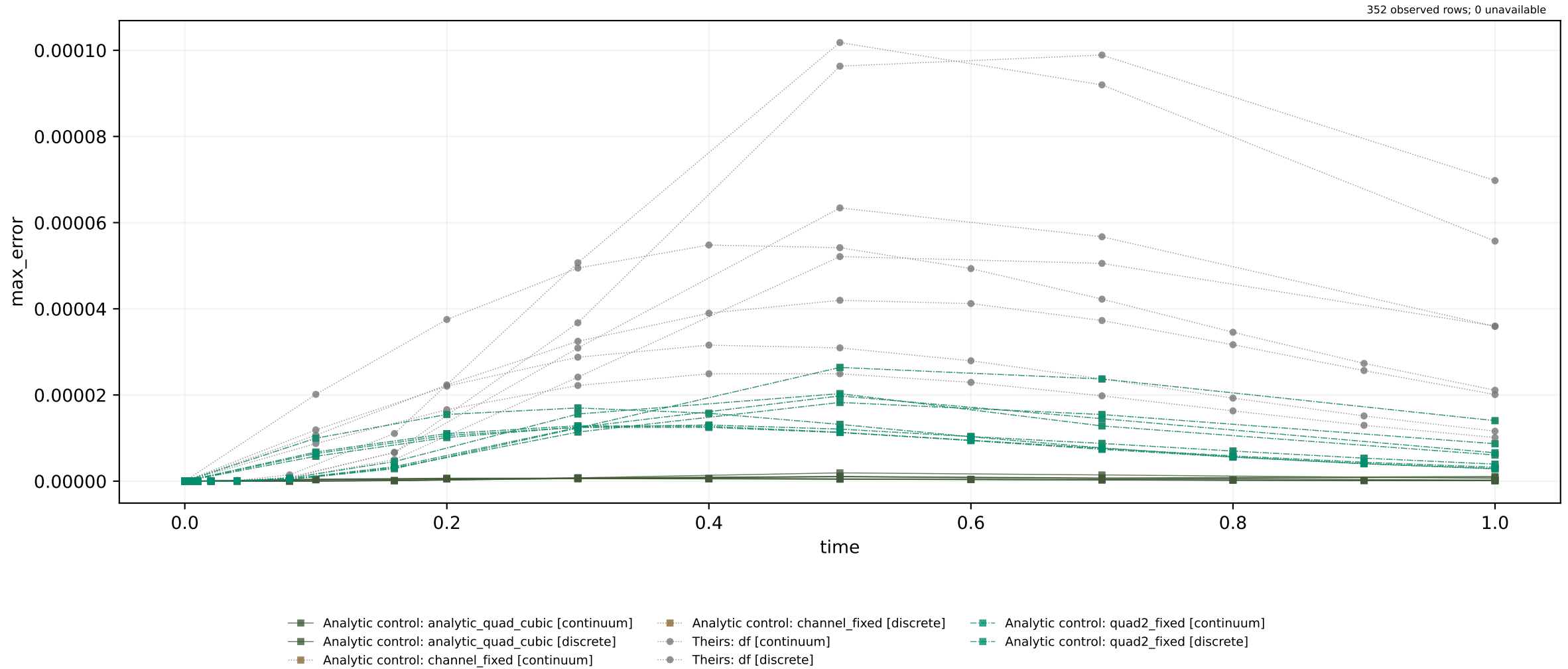
READ: Y closer to one means a larger cubic component orthogonal to the quadratic direction. That is new spatial information, not automatic accuracy or speed utility.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D06 Autonomous trajectory RMS



READ: Y lower is better throughout the trajectory. Own-output rollouts and teacher-forced one-step measurements are distinct; final equilibrium is insufficient.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D06 Worst cell trajectory error



READ: Lower Y is better. Maximum error can regress despite lower RMS; zero-time exactness is retained.

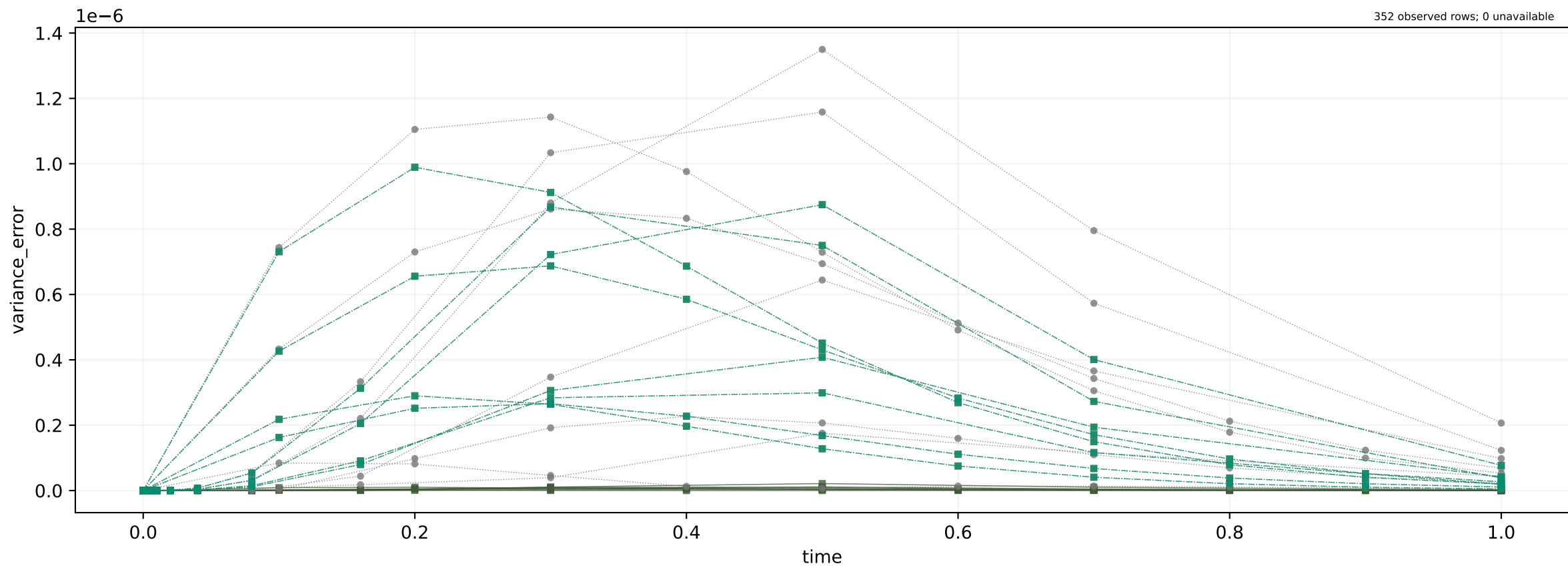
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D06 Variance trajectory error



READ: Lower absolute variance error is better at each matched time. Diffusive smoothing can erase initial roughness and hide transient defects.

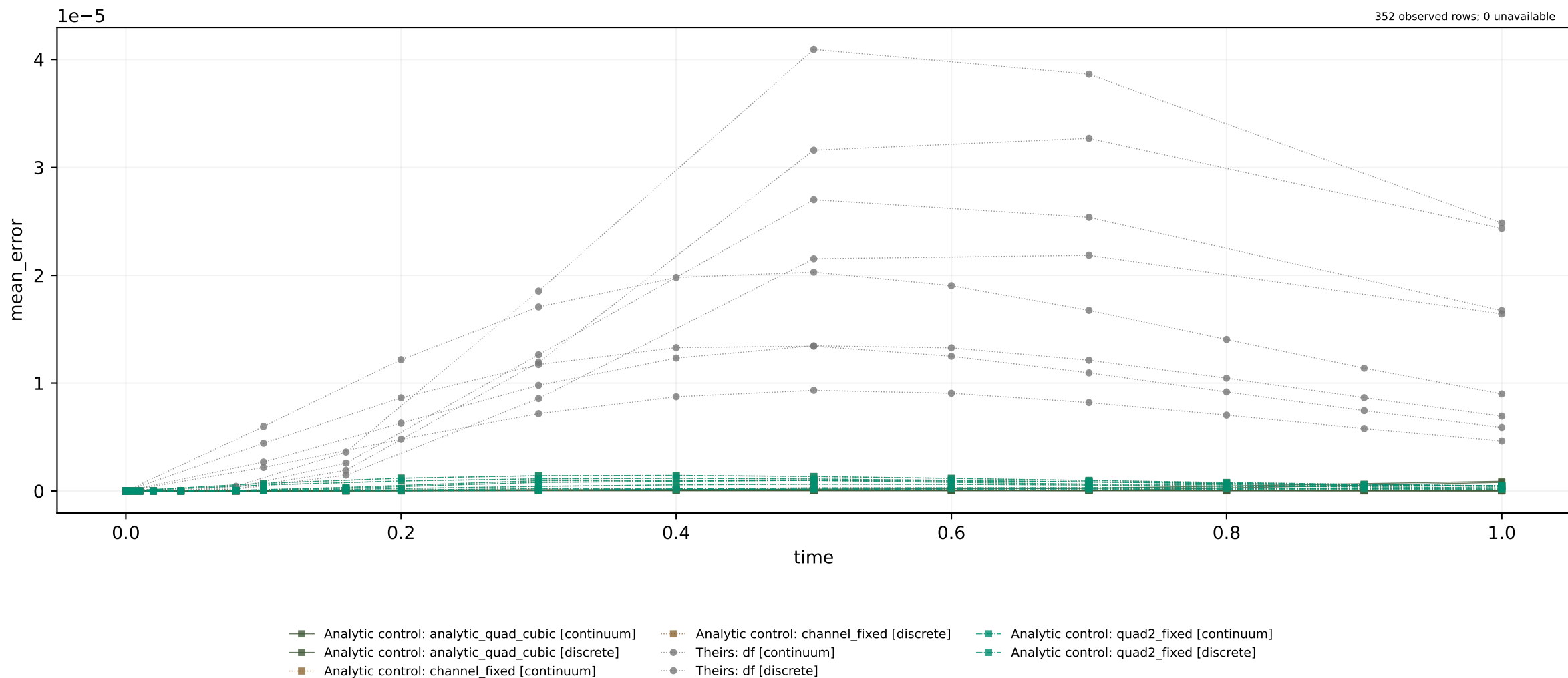
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D06 Mean variance and physical range



READ: Smaller absolute mean error is better. Logistic reaction does not conserve mean. Variance, range violations and late absolute error remain raw metrics.

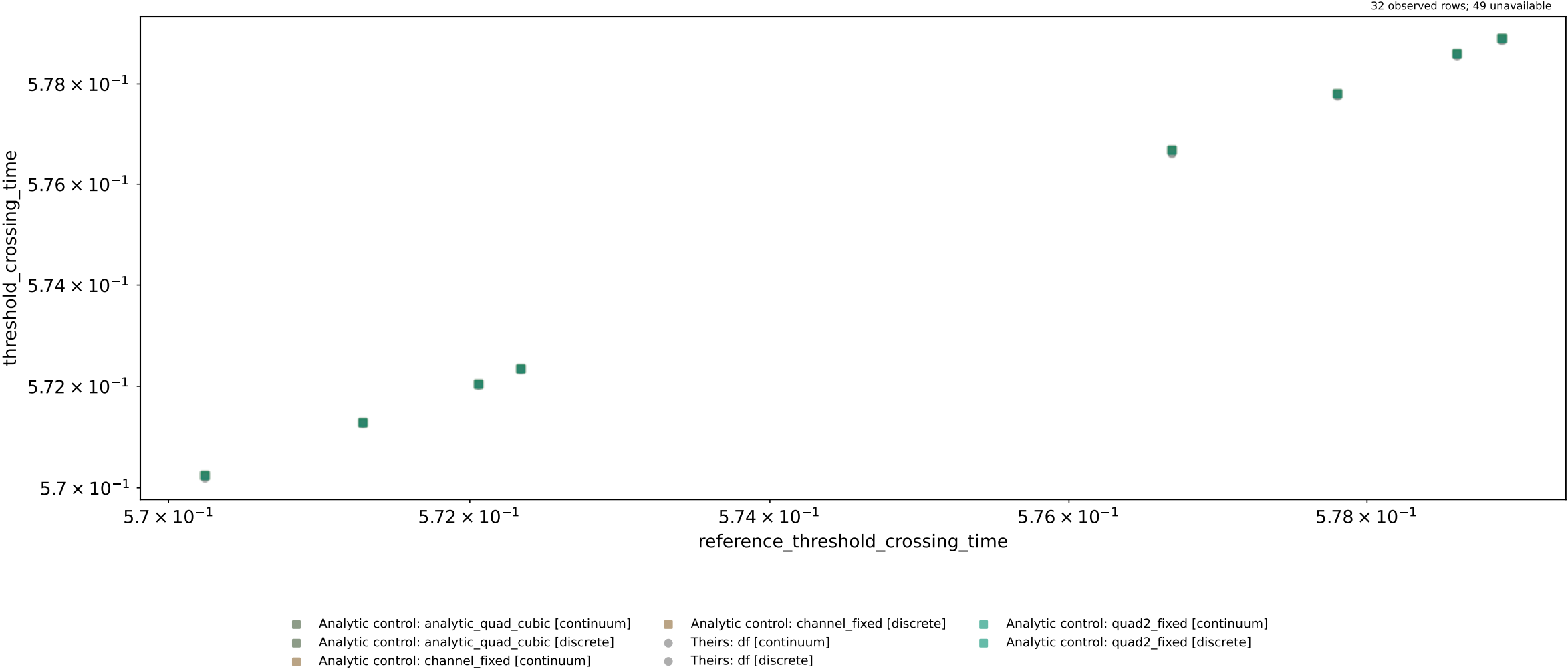
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D06 Threshold crossing times



READ: Agreement with the diagonal is better. Neither early nor late crossing is automatically accurate. Missing crossings remain unresolved.

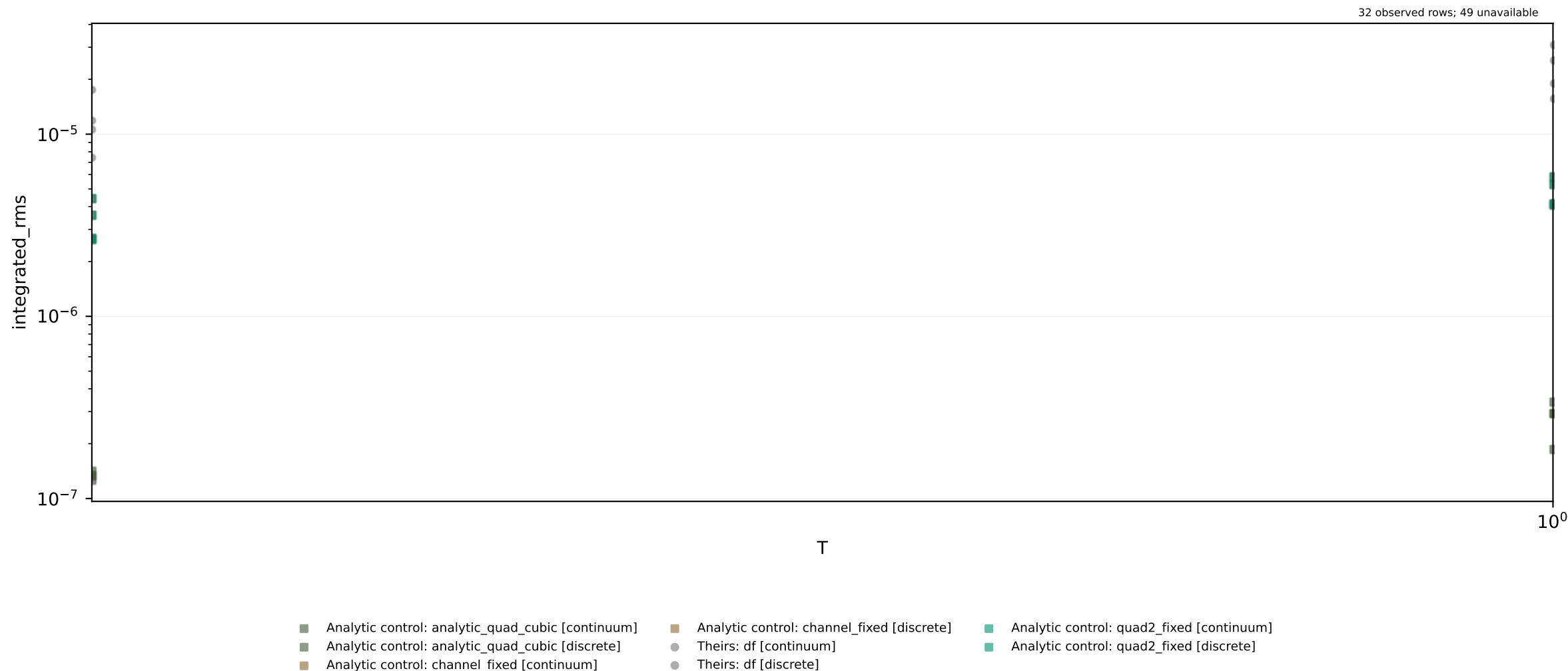
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

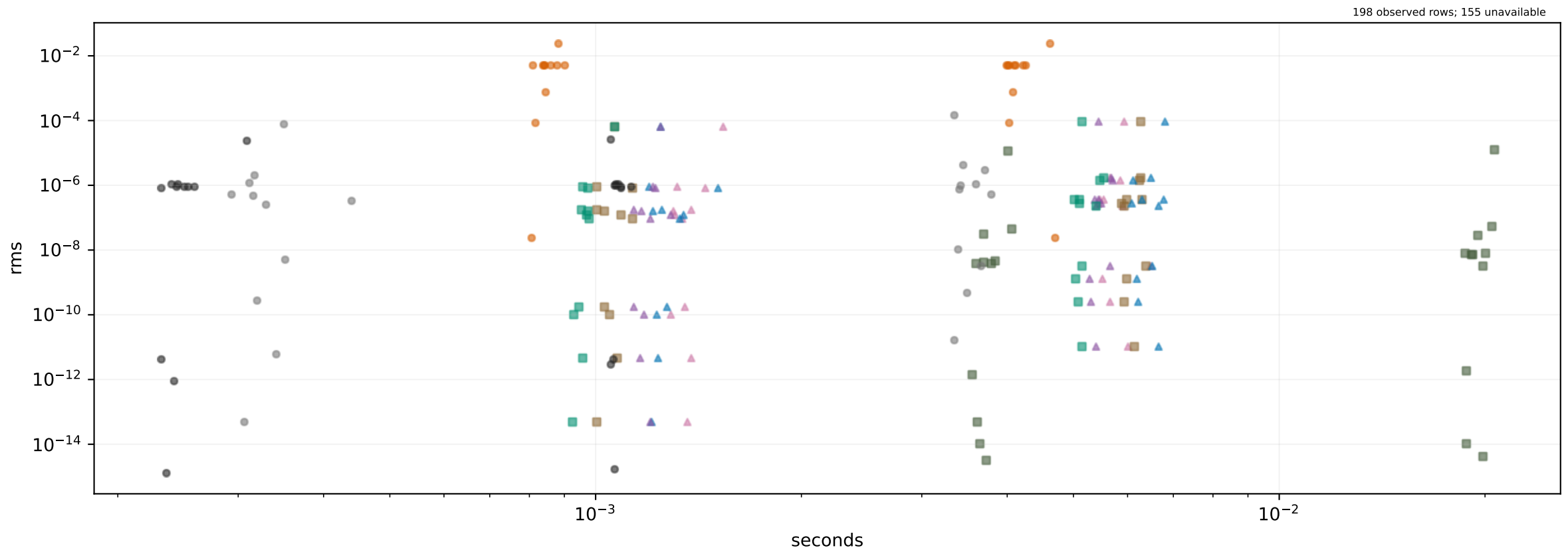
Missing evidence is NA.

# D06 Integrated error and stability



READ: Y lower is better over the same physical interval. Stable and accurate, stable but inaccurate, and unstable cases require separate denominators.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# D08 Competitor favorable regimes



READ: Lower-left is favorable within a matched regime. Smooth, weak-interaction and small-step workloads stay visible; short undertraining cannot establish neural inferiority.

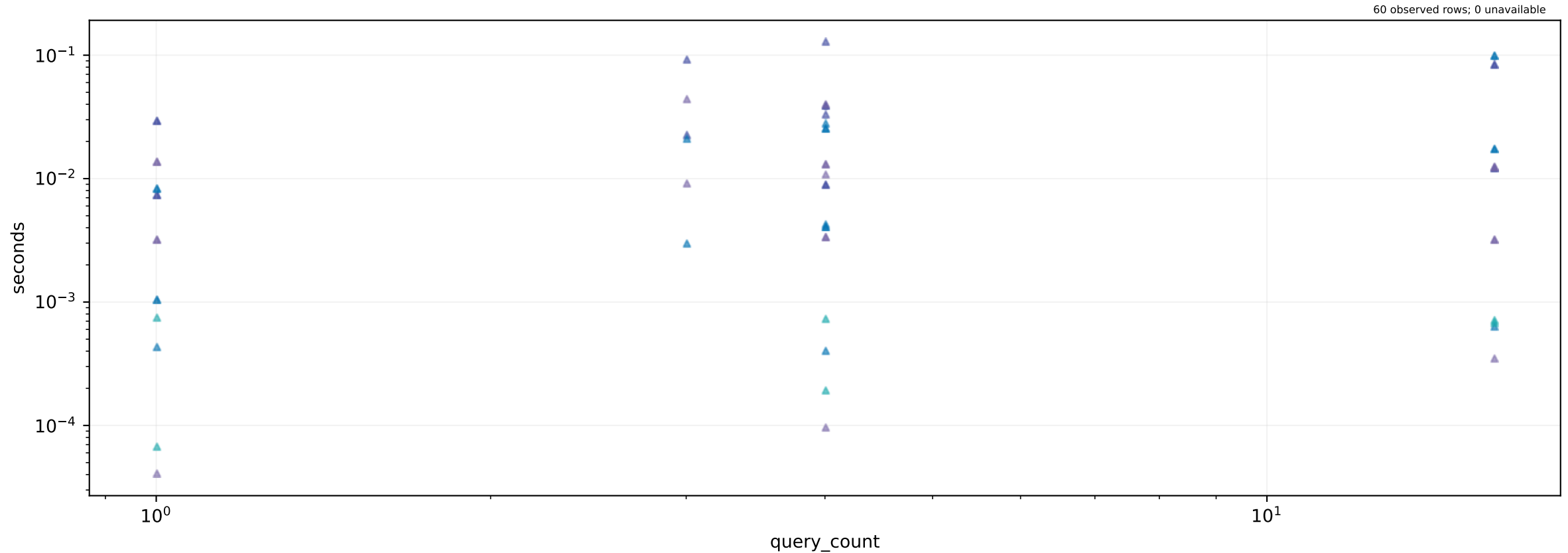
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

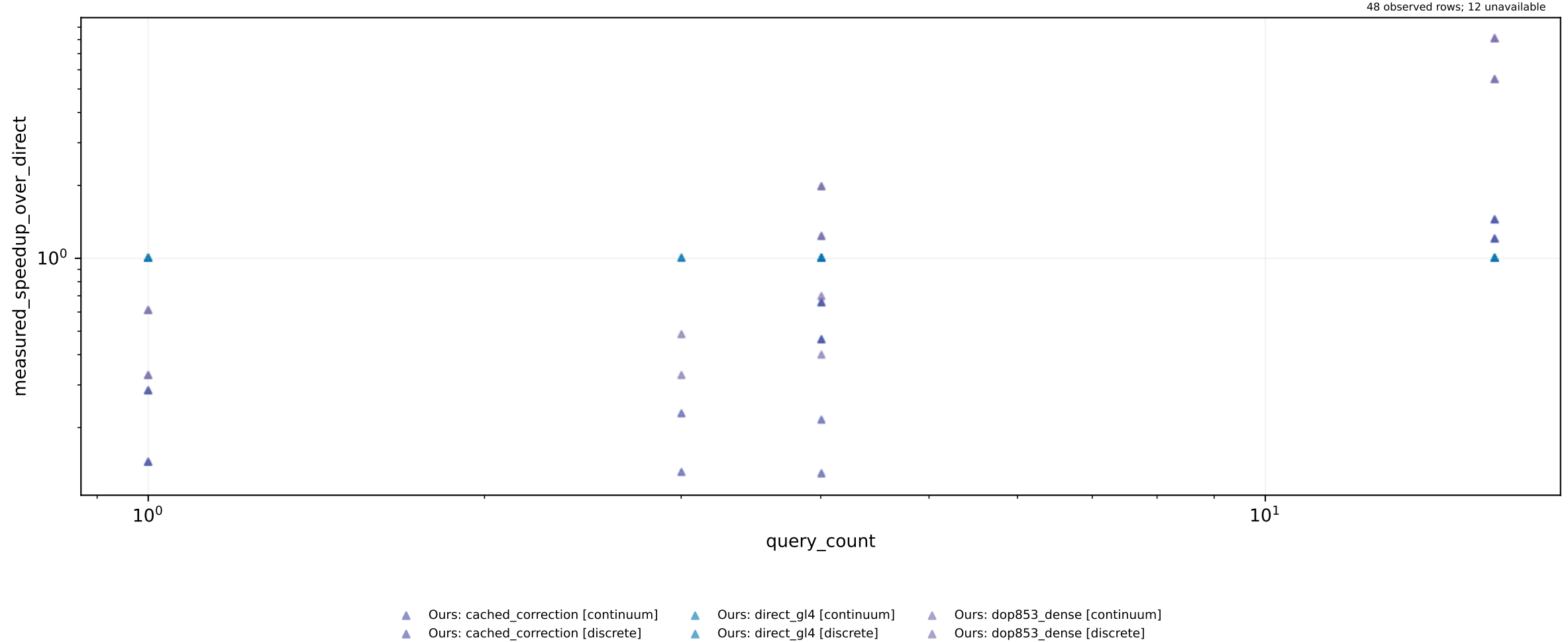
# D07 Actual multi query costs



- ▲ Ours: cached\_correction [continuum]
- ▲ Ours: cached\_correction [discrete]
- ▲ Ours: cached\_exact\_kernel
- ▲ Ours: compact\_kernel
- ▲ Ours: direct\_gl4 [continuum]
- ▲ Ours: direct\_gl4 [discrete]
- ▲ Ours: dop853\_dense [continuum]
- ▲ Ours: dop853\_dense [discrete]
- ▲ Ours: dop853\_lifted\_dense
- ▲ Ours: fft\_gl4

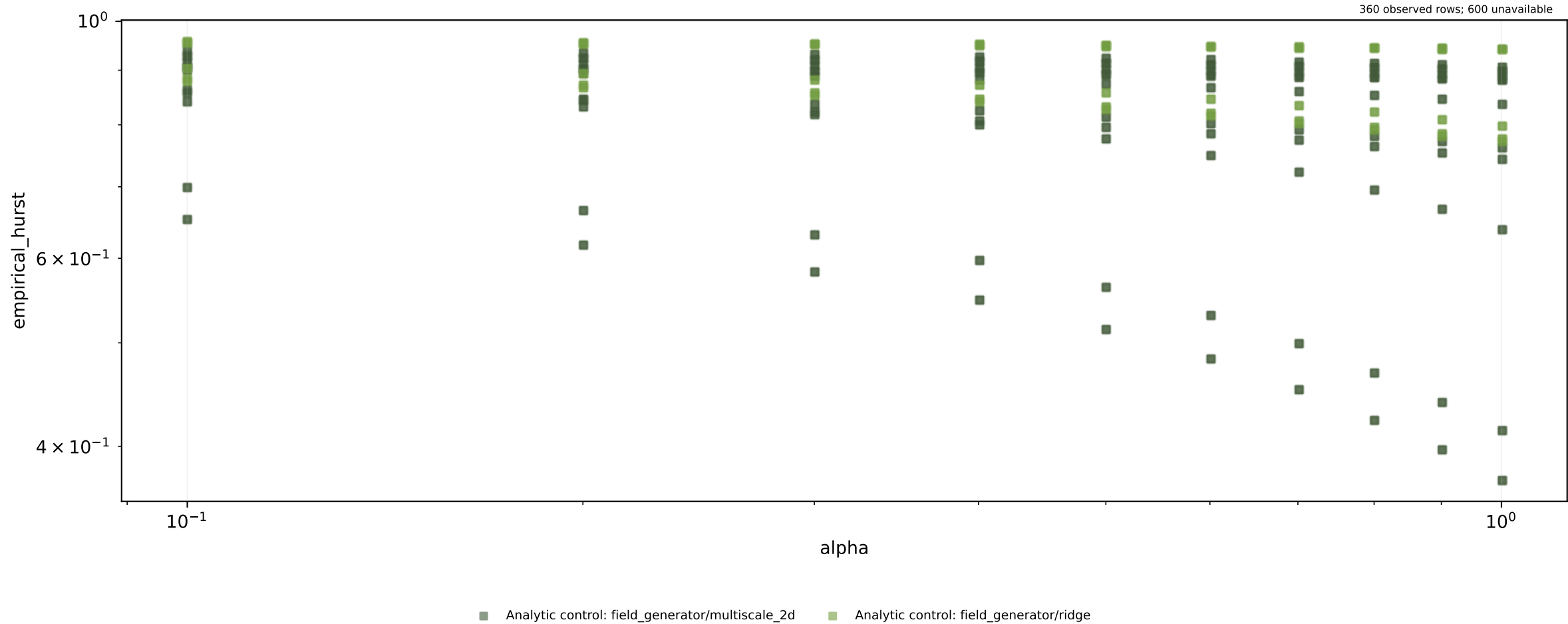
READ: Lower Y is better for the same accurate query workload. Compare measured totals with build plus Q times query models; caching must be equally available.  
 METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
 SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
 Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
 Missing evidence is NA.

# D07 Encoding refresh and break even



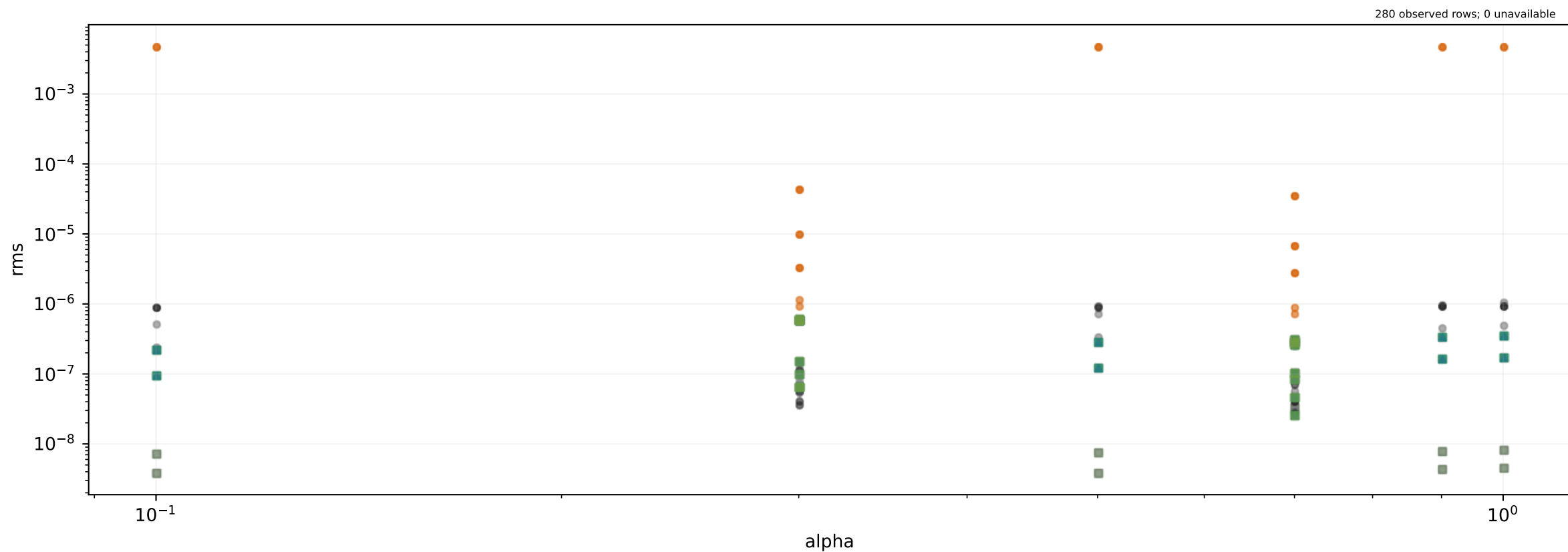
READ: Y greater than one favors the named reusable representation. No positive query saving means no finite break-even. Refresh and operator preparation are charged.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# Roughness approximation and empirical scaling



READ: X is a generator label, not a neural-network dimension. Empirical H is descriptive over the recorded scale range; finite trigonometric fields are smooth.  
 METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
 SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
 Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
 Missing evidence is NA.

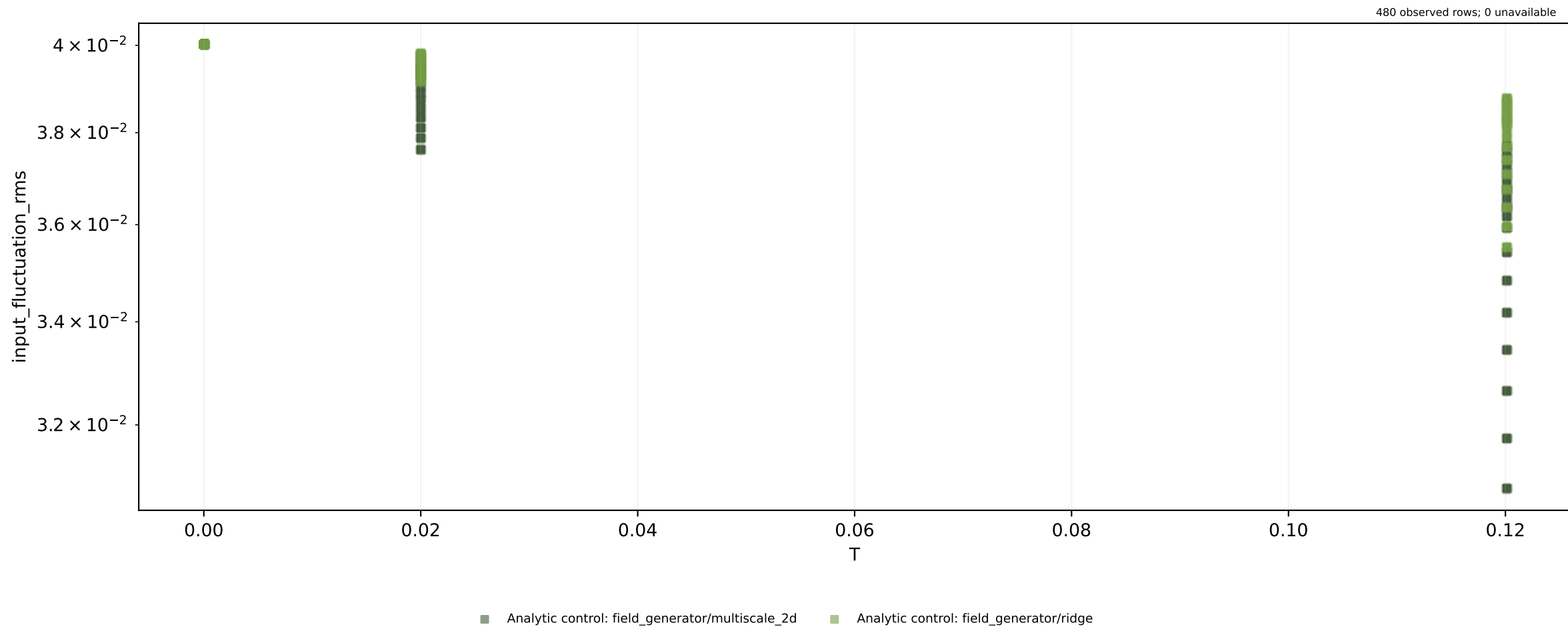
# Roughness accuracy and nonlinear interaction



- |                                                     |                                      |                                             |
|-----------------------------------------------------|--------------------------------------|---------------------------------------------|
| ■ Analytic control: analytic_quad_cubic [continuum] | ▲ Ours: channel_global [discrete]    | ● Theirs: fno_small [continuum]             |
| ■ Analytic control: analytic_quad_cubic [discrete]  | ▲ Ours: channel_neural [continuum]   | ● Theirs: fno_small [discrete]              |
| ▲ Ours: band_gain [continuum]                       | ▲ Ours: channel_neural [discrete]    | ▲ Ours: quad2_conditioned [continuum]       |
| ▲ Ours: band_gain [discrete]                        | ● Theirs: df [continuum]             | ▲ Ours: quad2_conditioned [discrete]        |
| ▲ Ours: channel_affine [continuum]                  | ● Theirs: df [discrete]              | ■ Analytic control: quad2_fixed [continuum] |
| ▲ Ours: channel_affine [discrete]                   | ● Theirs: etdrk4 [continuum]         | ■ Analytic control: quad2_fixed [discrete]  |
| ■ Analytic control: channel_fixed [continuum]       | ● Theirs: etdrk4 [discrete]          | ■ Analytic control: quad4_fixed [continuum] |
| ■ Analytic control: channel_fixed [discrete]        | ▲ Ours: feature_capacity [continuum] | ■ Analytic control: quad4_fixed [discrete]  |
| ▲ Ours: channel_global [continuum]                  | ▲ Ours: feature_capacity [discrete]  |                                             |

READ: Lower Y is better at fixed mean, amplitude, bandwidth, grid and horizon. Alpha=1 is a finite-bandwidth stress endpoint, not the convergent ridge theorem.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# Roughness heat-only amplitude evolution



READ: Y is fluctuation amplitude after exact heat-only smoothing, not error and not full reaction-diffusion. Lower amplitude means smoother/weaker fluctuations; it is not a model win. X is physical time.

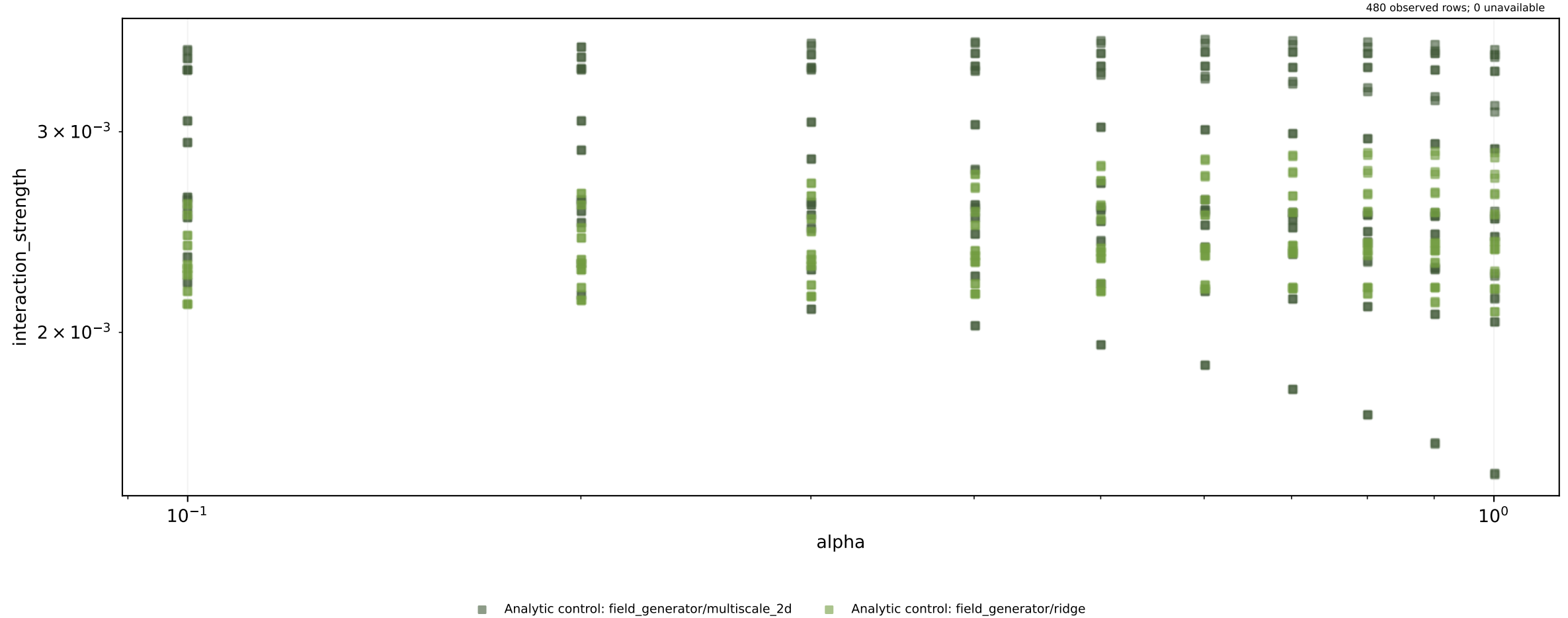
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# Roughness signed-product strength



READ: Y measures nodal centered-square RMS, not solver error or a dealiased Galerkin defect. Neither higher nor lower is automatically better. X is the ideal-generator label; finite samples are smooth.

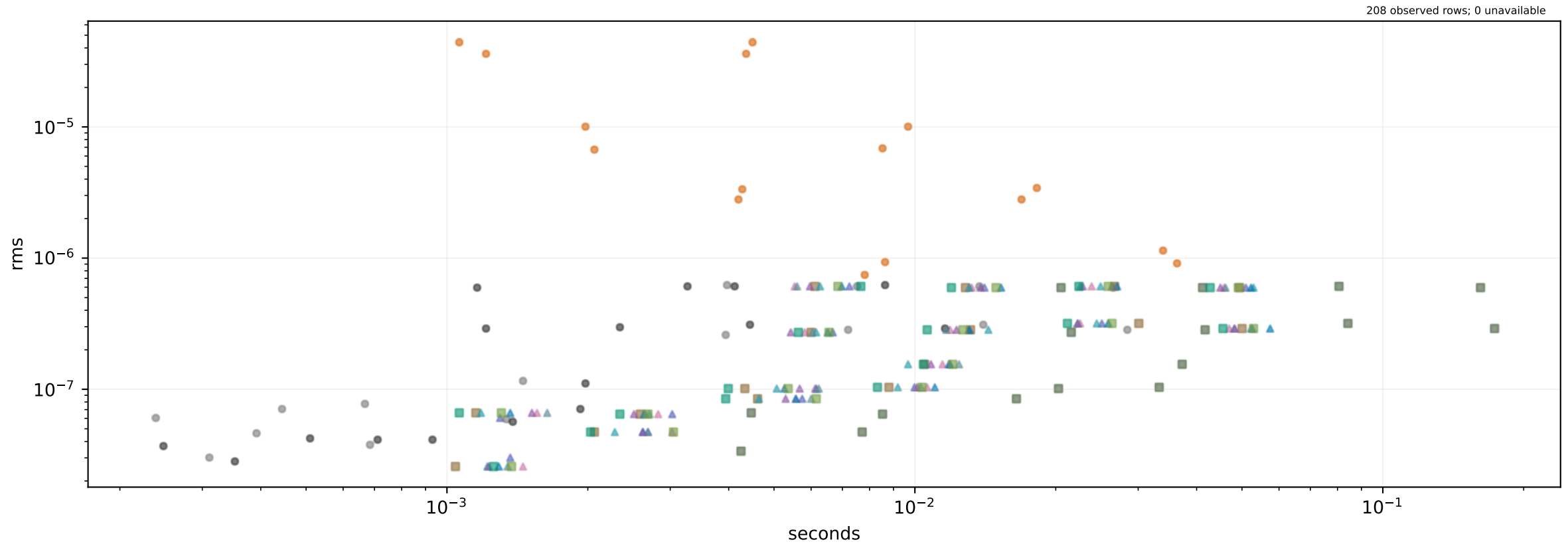
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# Pilot accuracy cost comparisons



- |                                                     |                                      |                                             |
|-----------------------------------------------------|--------------------------------------|---------------------------------------------|
| ■ Analytic control: analytic_quad_cubic [continuum] | ▲ Ours: channel_global [discrete]    | ● Theirs: fno_small [continuum]             |
| ■ Analytic control: analytic_quad_cubic [discrete]  | ▲ Ours: channel_neural [continuum]   | ● Theirs: fno_small [discrete]              |
| ▲ Ours: band_gain [continuum]                       | ▲ Ours: channel_neural [discrete]    | ▲ Ours: quad2_conditioned [continuum]       |
| ▲ Ours: band_gain [discrete]                        | ● Theirs: df [continuum]             | ▲ Ours: quad2_conditioned [discrete]        |
| ▲ Ours: channel_affine [continuum]                  | ● Theirs: df [discrete]              | ■ Analytic control: quad2_fixed [continuum] |
| ▲ Ours: channel_affine [discrete]                   | ● Theirs: etdrk4 [continuum]         | ■ Analytic control: quad2_fixed [discrete]  |
| ■ Analytic control: channel_fixed [continuum]       | ● Theirs: etdrk4 [discrete]          | ■ Analytic control: quad4_fixed [continuum] |
| ■ Analytic control: channel_fixed [discrete]        | ▲ Ours: feature_capacity [continuum] | ■ Analytic control: quad4_fixed [discrete]  |
| ▲ Ours: channel_global [continuum]                  | ▲ Ours: feature_capacity [discrete]  |                                             |

READ: Lower-left is favorable only at matched accuracy definitions and physical tasks. These are development pilots unless a fresh locked confirmation is explicitly supplied.

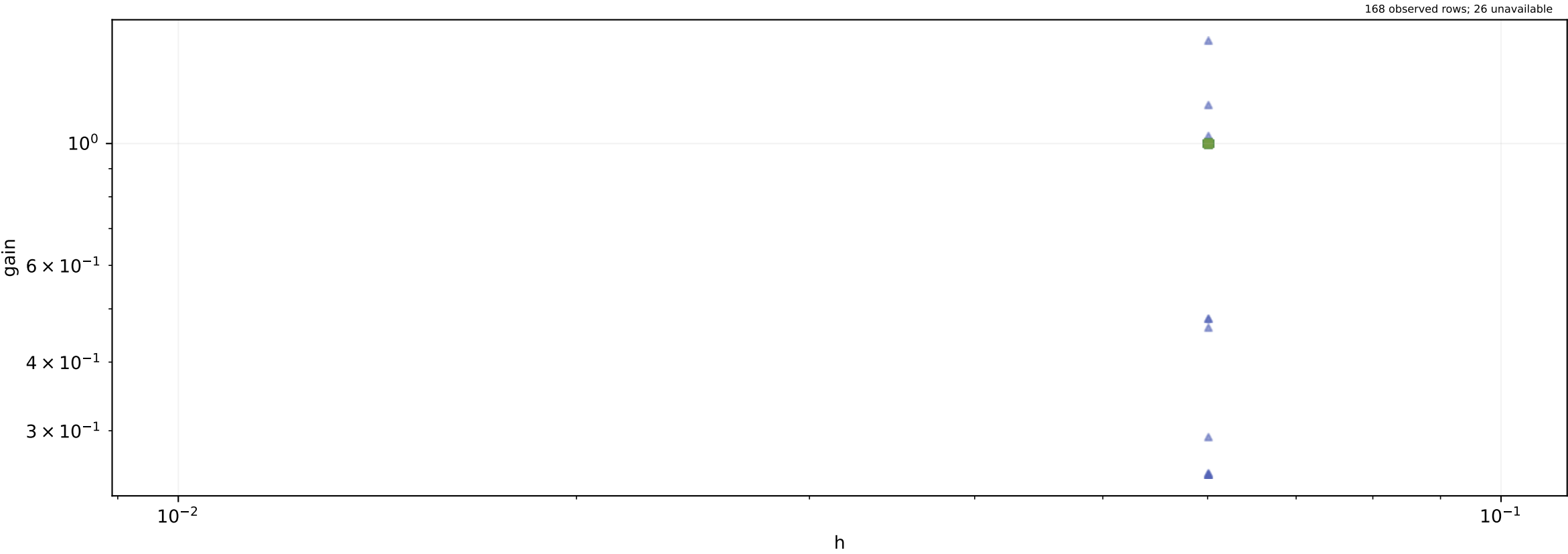
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

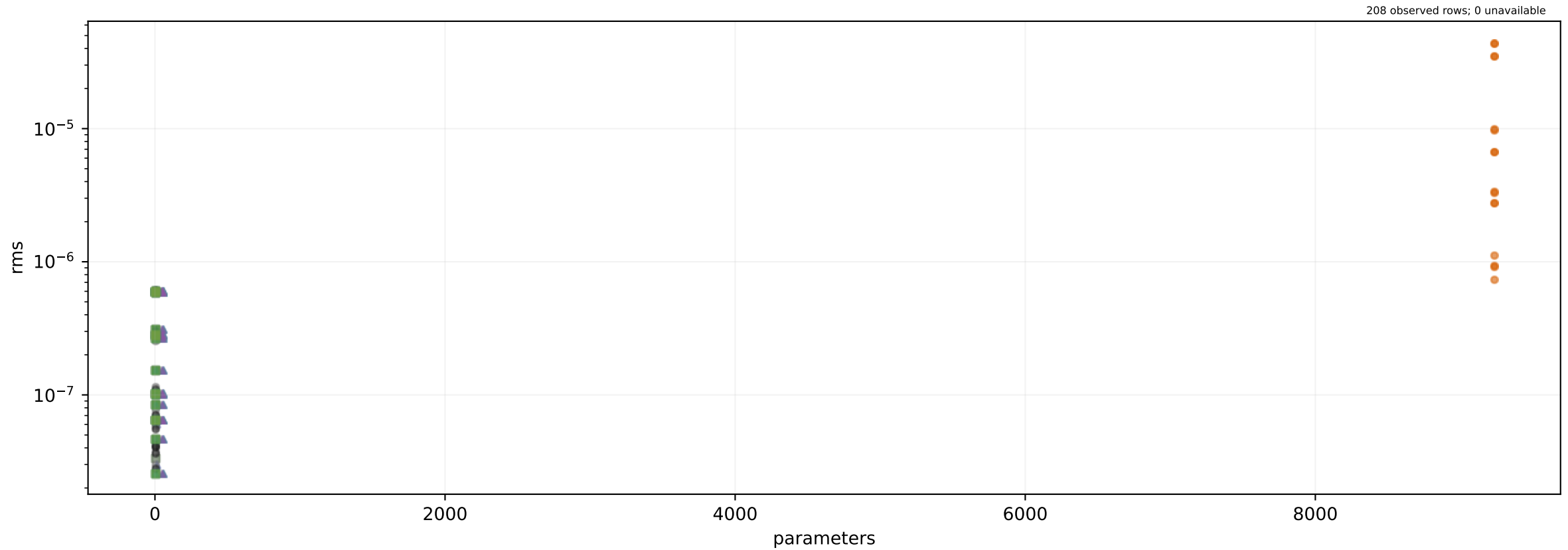
# Pilot gain responses



- |                                               |                                      |                                             |
|-----------------------------------------------|--------------------------------------|---------------------------------------------|
| ▲ Ours: band_gain [continuum]                 | ▲ Ours: channel_global [continuum]   | ▲ Ours: quad2_conditioned [continuum]       |
| ▲ Ours: band_gain [discrete]                  | ▲ Ours: channel_global [discrete]    | ▲ Ours: quad2_conditioned [discrete]        |
| ▲ Ours: channel_affine [continuum]            | ▲ Ours: channel_neural [continuum]   | ■ Analytic control: quad2_fixed [continuum] |
| ▲ Ours: channel_affine [discrete]             | ▲ Ours: channel_neural [discrete]    | ■ Analytic control: quad2_fixed [discrete]  |
| ■ Analytic control: channel_fixed [continuum] | ▲ Ours: feature_capacity [continuum] | ■ Analytic control: quad4_fixed [continuum] |
| ■ Analytic control: channel_fixed [discrete]  | ▲ Ours: feature_capacity [discrete]  | ■ Analytic control: quad4_fixed [discrete]  |

READ: X is step size; neither larger nor smaller gain is inherently better. Inspect response identifiability and sensitivity before attributing physical meaning.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# Pilot model size and error



- |                                                     |                                      |                                             |
|-----------------------------------------------------|--------------------------------------|---------------------------------------------|
| ■ Analytic control: analytic_quad_cubic [continuum] | ▲ Ours: channel_global [discrete]    | ● Theirs: fno_small [continuum]             |
| ■ Analytic control: analytic_quad_cubic [discrete]  | ▲ Ours: channel_neural [continuum]   | ● Theirs: fno_small [discrete]              |
| ▲ Ours: band_gain [continuum]                       | ▲ Ours: channel_neural [discrete]    | ▲ Ours: quad2_conditioned [continuum]       |
| ▲ Ours: band_gain [discrete]                        | ● Theirs: df [continuum]             | ▲ Ours: quad2_conditioned [discrete]        |
| ▲ Ours: channel_affine [continuum]                  | ● Theirs: df [discrete]              | ■ Analytic control: quad2_fixed [continuum] |
| ▲ Ours: channel_affine [discrete]                   | ● Theirs: etdrk4 [continuum]         | ■ Analytic control: quad2_fixed [discrete]  |
| ■ Analytic control: channel_fixed [continuum]       | ● Theirs: etdrk4 [discrete]          | ■ Analytic control: quad4_fixed [continuum] |
| ■ Analytic control: channel_fixed [discrete]        | ▲ Ours: feature_capacity [continuum] | ■ Analytic control: quad4_fixed [discrete]  |
| ▲ Ours: channel_global [continuum]                  | ▲ Ours: feature_capacity [discrete]  |                                             |

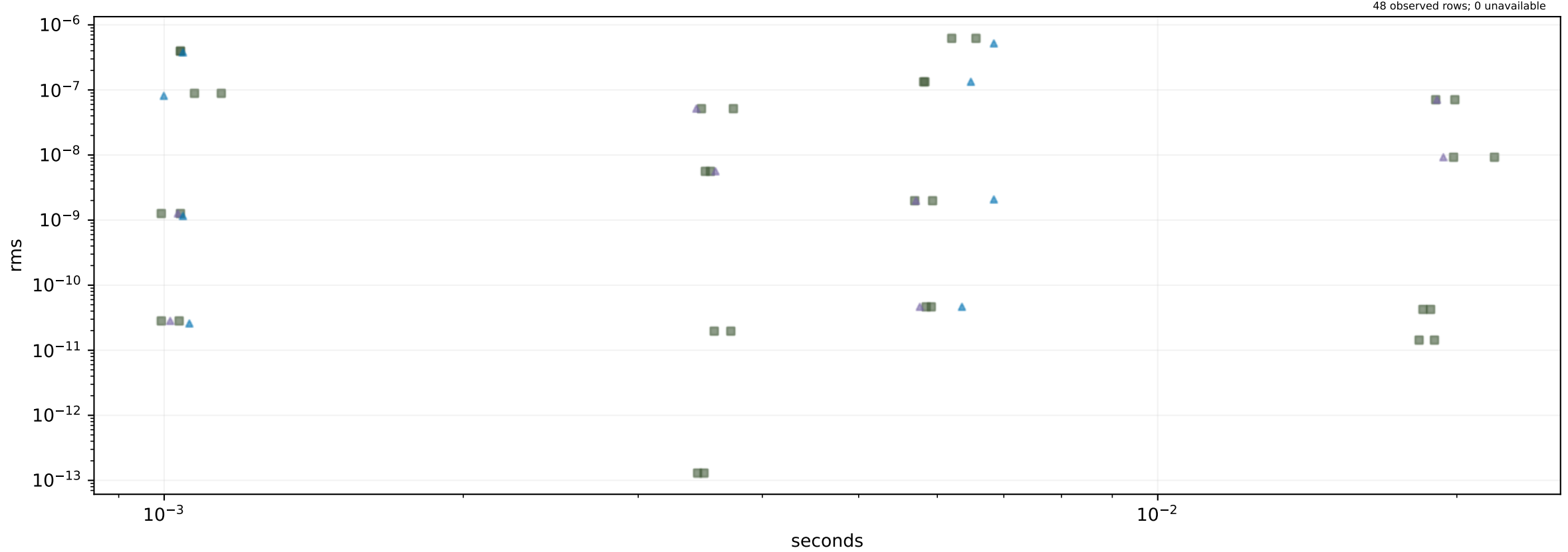
READ: Lower-left is compact and accurate; parameter count is not measured compute or a fairness definition. Fixed and initialized selections remain visible.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

Peak measured memory

NA: no verified measurements for this panel

READ: Lower Y uses less GPU memory for matched batch, precision and work. Host RAM is not VRAM; missing GPU measurements remain NA.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# Exploratory alternatives



READ: Lower-left is favorable for matched targets. A successful mathematical prototype is not a complete solver or evidence of useful amortization.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

## D06 Stability classifications and denominators



READ: Accurate/stable is favorable; stable/inaccurate and unstable remain distinct. Counts refer to the declared paired rollouts, not independent samples or confidence bounds.

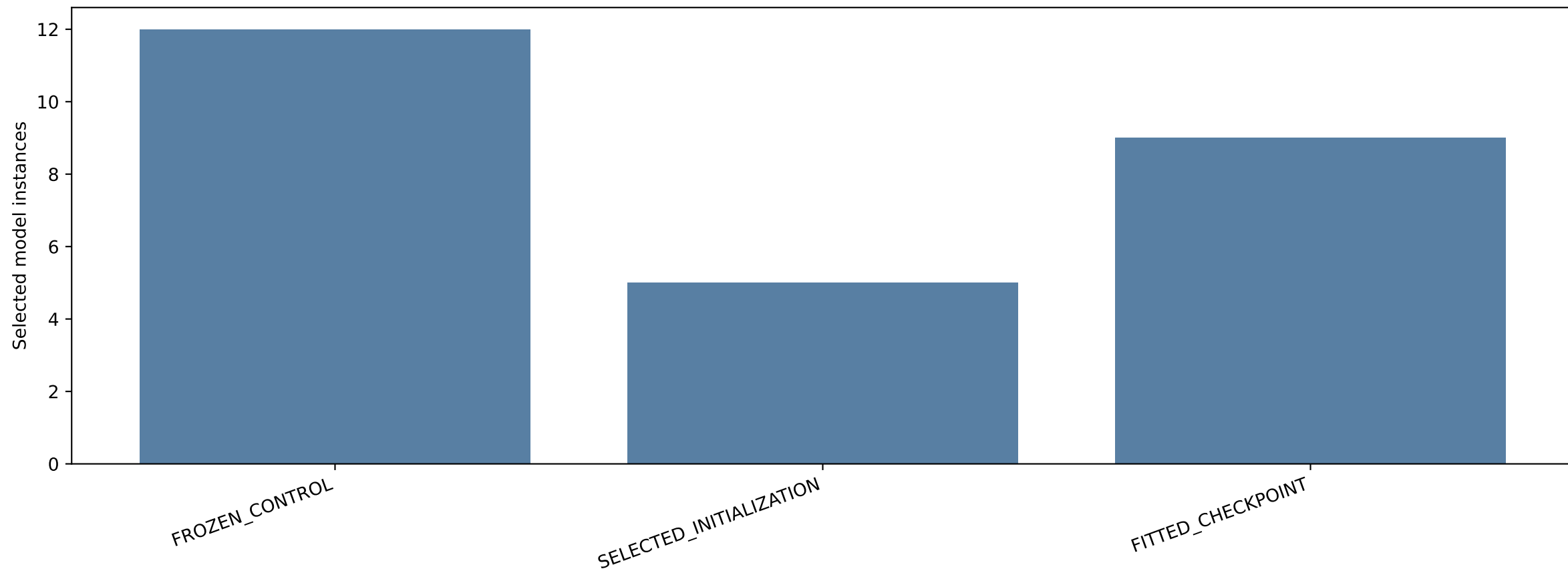
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## Pilot checkpoint and baseline adequacy



READ: A fitted selection shows that this validation objective improved over initialization; it does not establish optimization convergence. Initialization-selected or failed competitors leave broad inferiority claims unresolved.

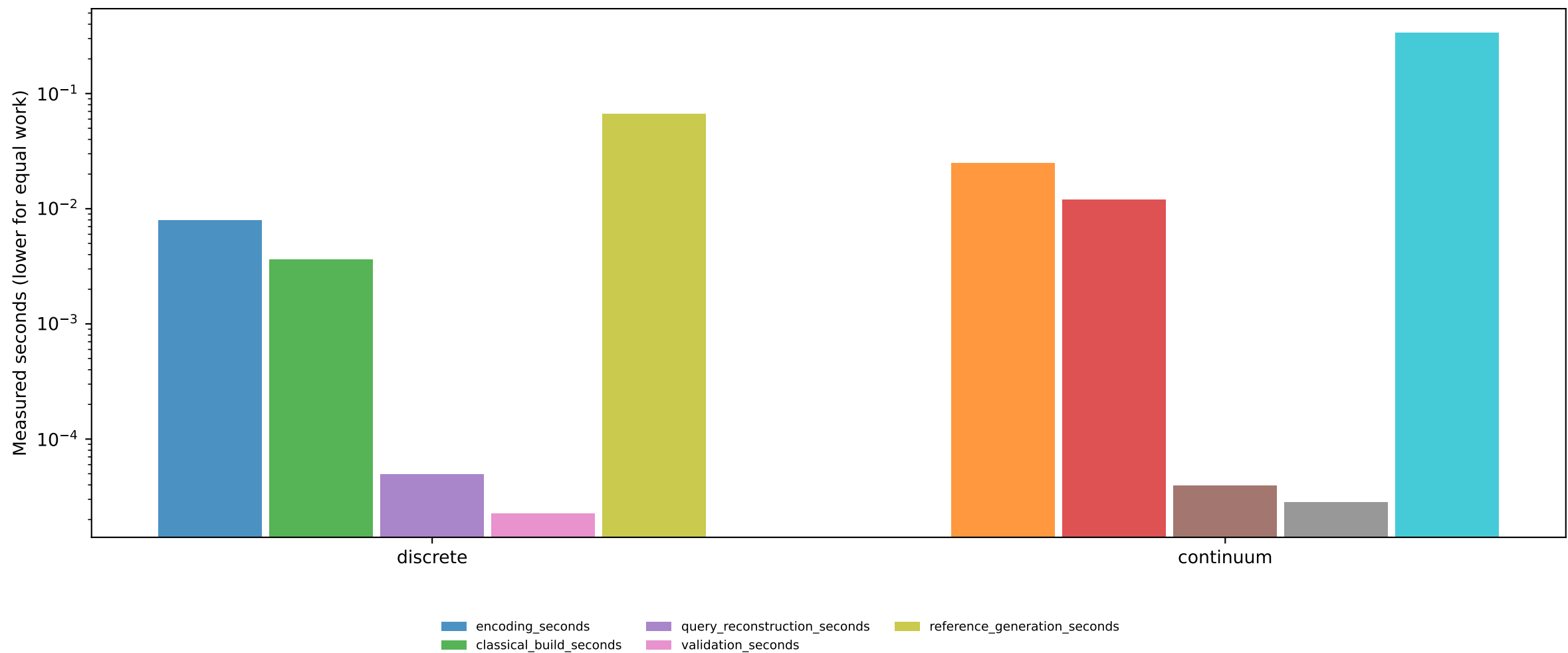
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

# D07 Encoding reference and reconstruction costs



READ: Lower seconds is cheaper for equal work. Reference generation is offline cost, while encoding, validation and reconstruction may be paid at runtime. Compilation and energy stay NA when unavailable.

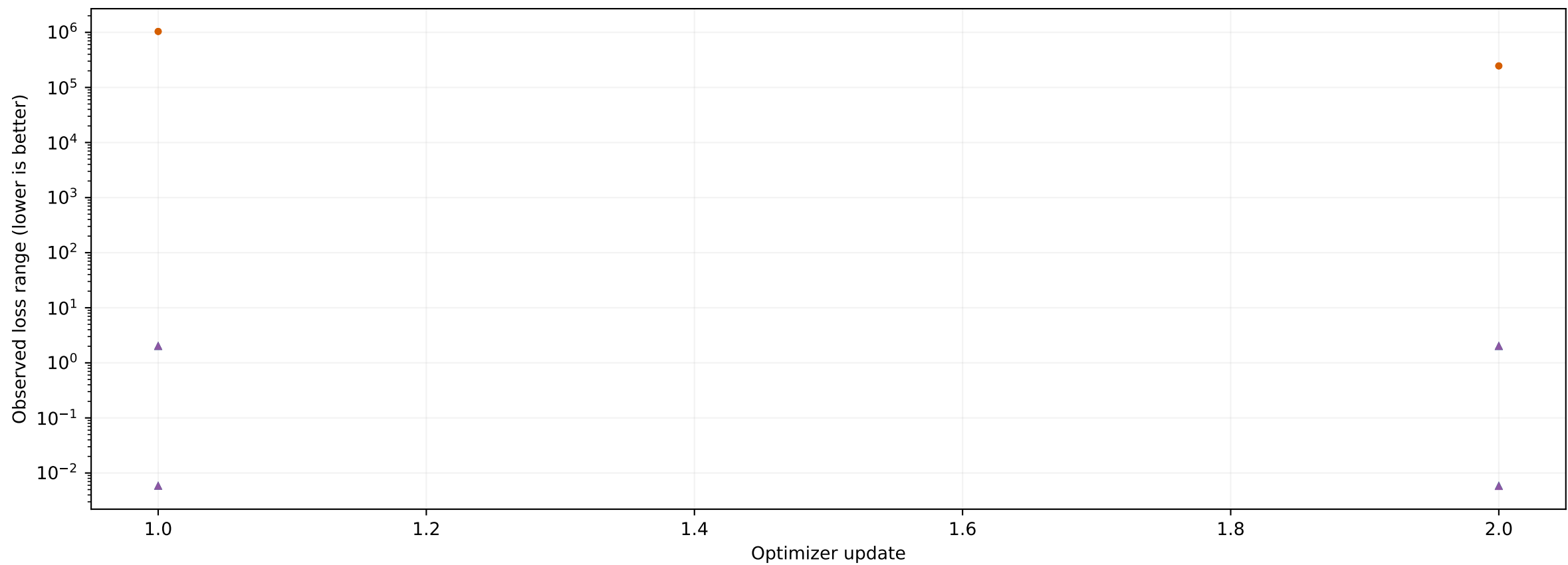
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

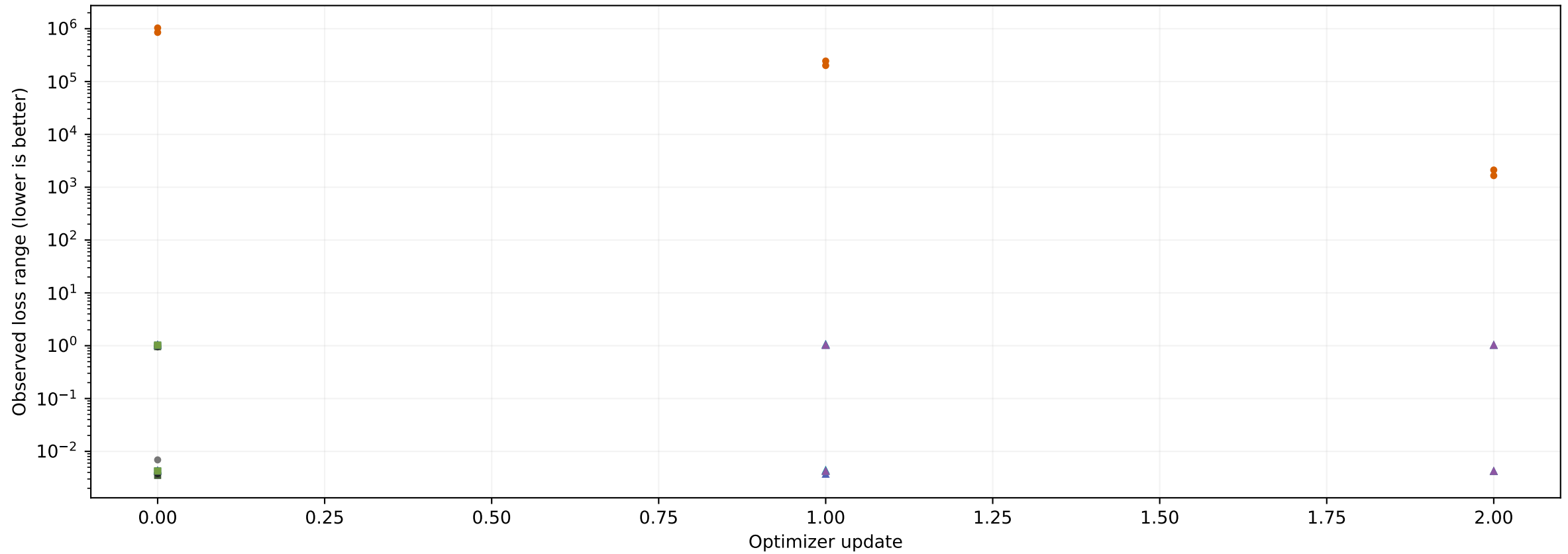
# Learning loss observed ranges



—▲— Ours: band\_gain [continuum, training]      —▲— Ours: feature\_capacity [continuum, training]      —●— Theirs: fno\_small [discrete, training]  
—▲— Ours: band\_gain [discrete, training]      —▲— Ours: feature\_capacity [discrete, training]      —▲— Ours: quad2\_conditioned [continuum, training]  
—▲— Ours: channel\_neural [continuum, training]      —●— Theirs: fno\_small [continuum, training]      —▲— Ours: quad2\_conditioned [discrete, training]  
—▲— Ours: channel\_neural [discrete, training]

READ: Lower loss is a closer fit to the stated training objective. Translucent bands span observed minima and maxima in adjacent windows; they are continuous observed ranges, not confidence intervals. Empty windows leave visible gaps.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.

# Validation loss observed ranges



READ: Lower validation loss is better on the same objective. Observed low/high ranges are not confidence intervals; unmeasured validation windows remain gaps. Selection does not replace fresh confirmation.

METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.

SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.

Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.

Missing evidence is NA.

## Complete cost ledger and limitations

Recorded scientific-stage time: 78.106 s across 14 measured units.

This is a partial ledger unless operational startup, preprocessing, references, tuning, failures, recovery, reporting and allocations are also supplied.

Energy: NA without measurements. Monetary cost: NA without an actual rate and accounting model.

Comparative claims require same-task accuracy-qualified full solver cost, including estimation, rejection and fallback where used.

READ: Smaller complete cost is better for the same accuracy and deployment volume. Missing charges are not zero; ceilings and operation counts are not resource bills.  
METHODS: Ours: triangles; Theirs: circles; analytic controls: squares. Thin line styles and colors distinguish methods. Local FNO controls are not paper reproductions.  
SCOPE: Adjacent study only; no promotion into the main architecture. Computational completion, mathematical checks and scientific utility are separate outcomes.  
Independent parents are the statistical units; paired grids, phases, schedules and seeds are repeated observations. Observed-range bands are not confidence intervals.  
Missing evidence is NA.